

BFS Traversal

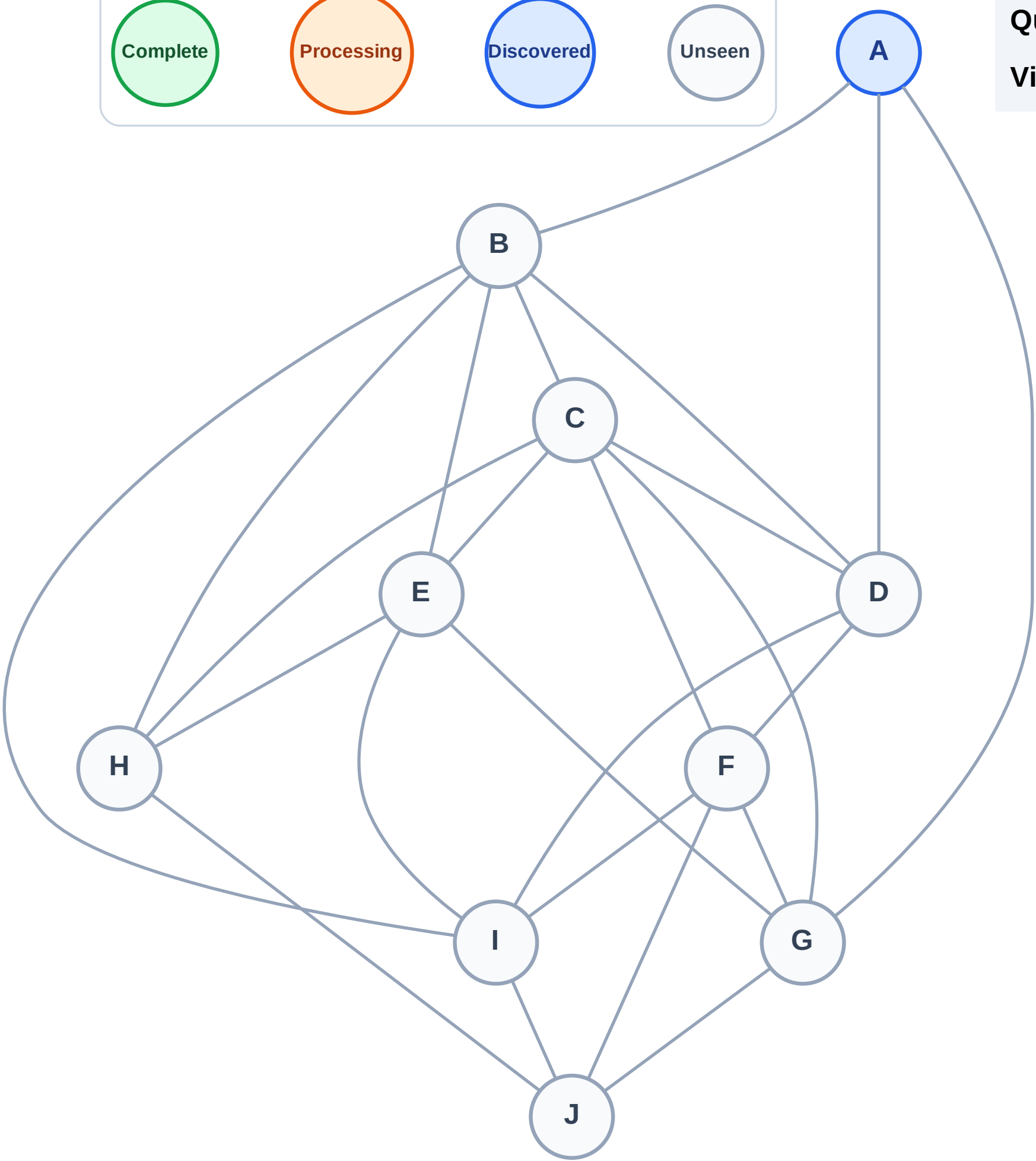
Step 1: Start at A

Legend



Queue: A

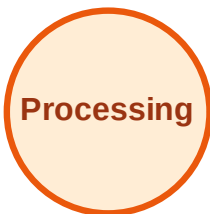
Visited: (none)



BFS Traversal

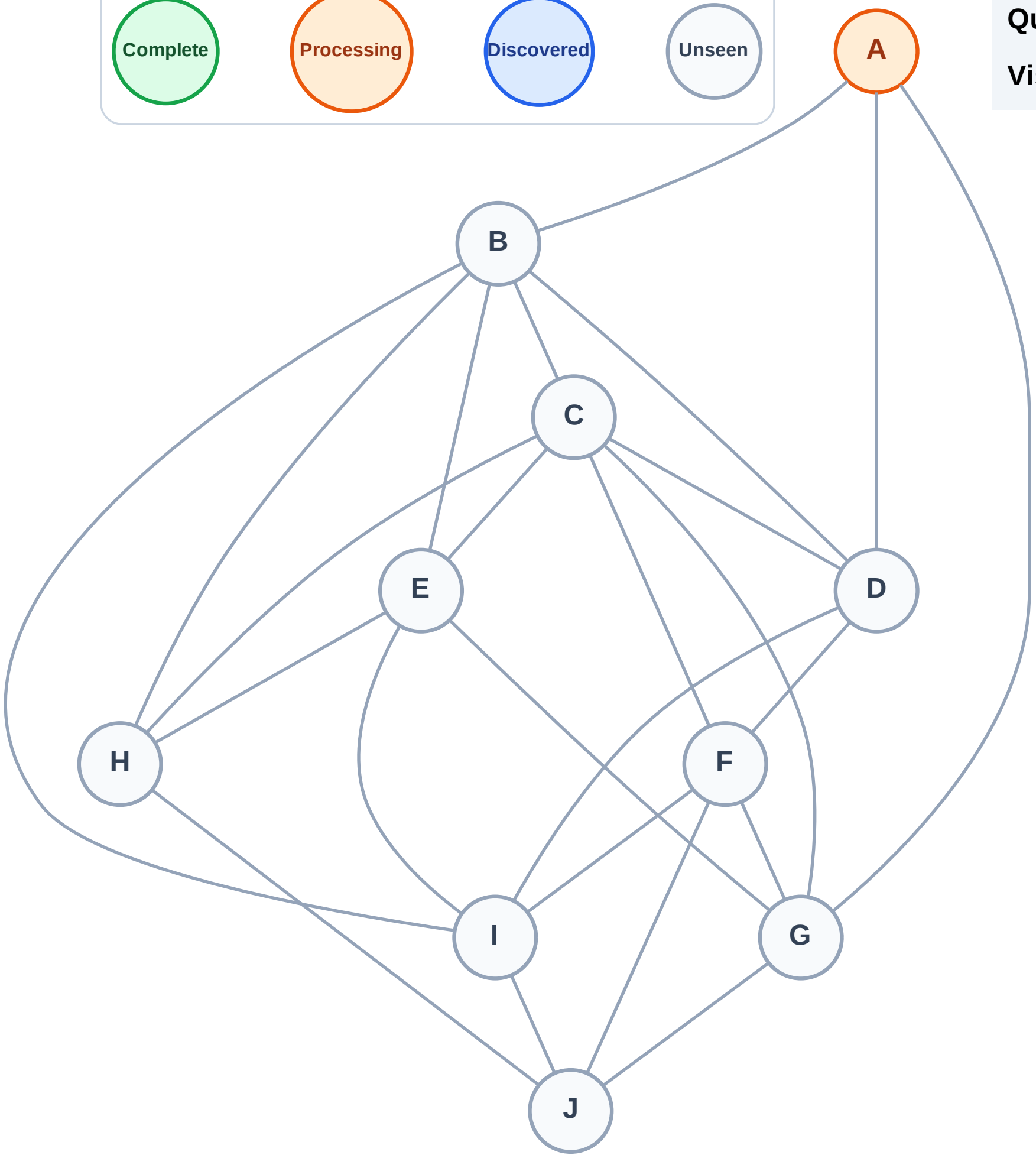
Step 2: Process node A

Legend



Queue: (empty)

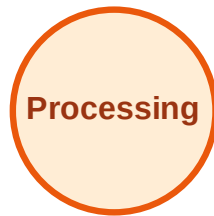
Visited: (none)



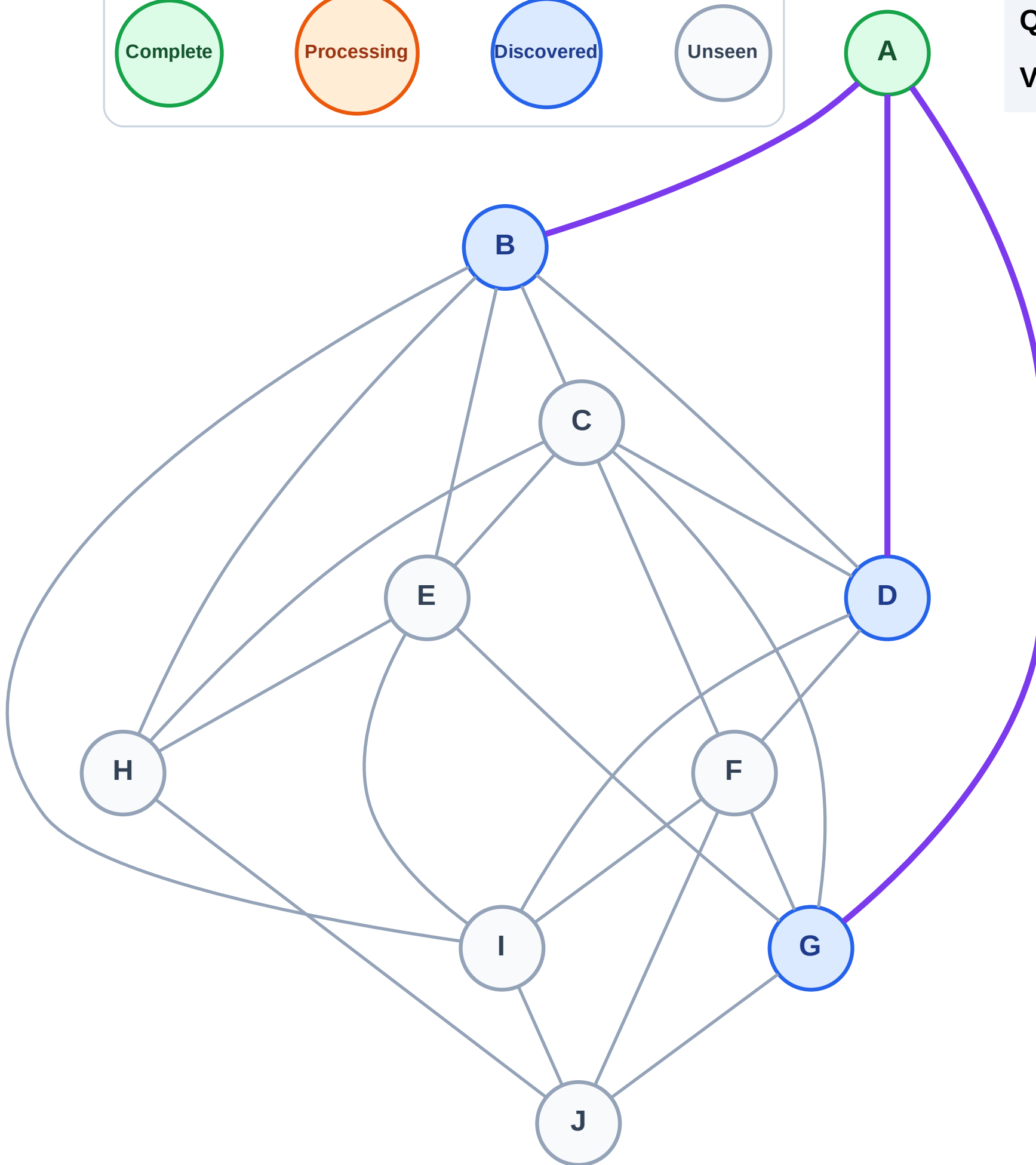
BFS Traversal

Step 3: Discover B, D, G from A

Legend



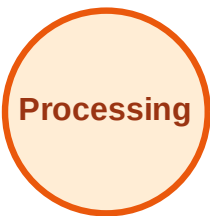
Queue: B → D → G
Visited: A



BFS Traversal

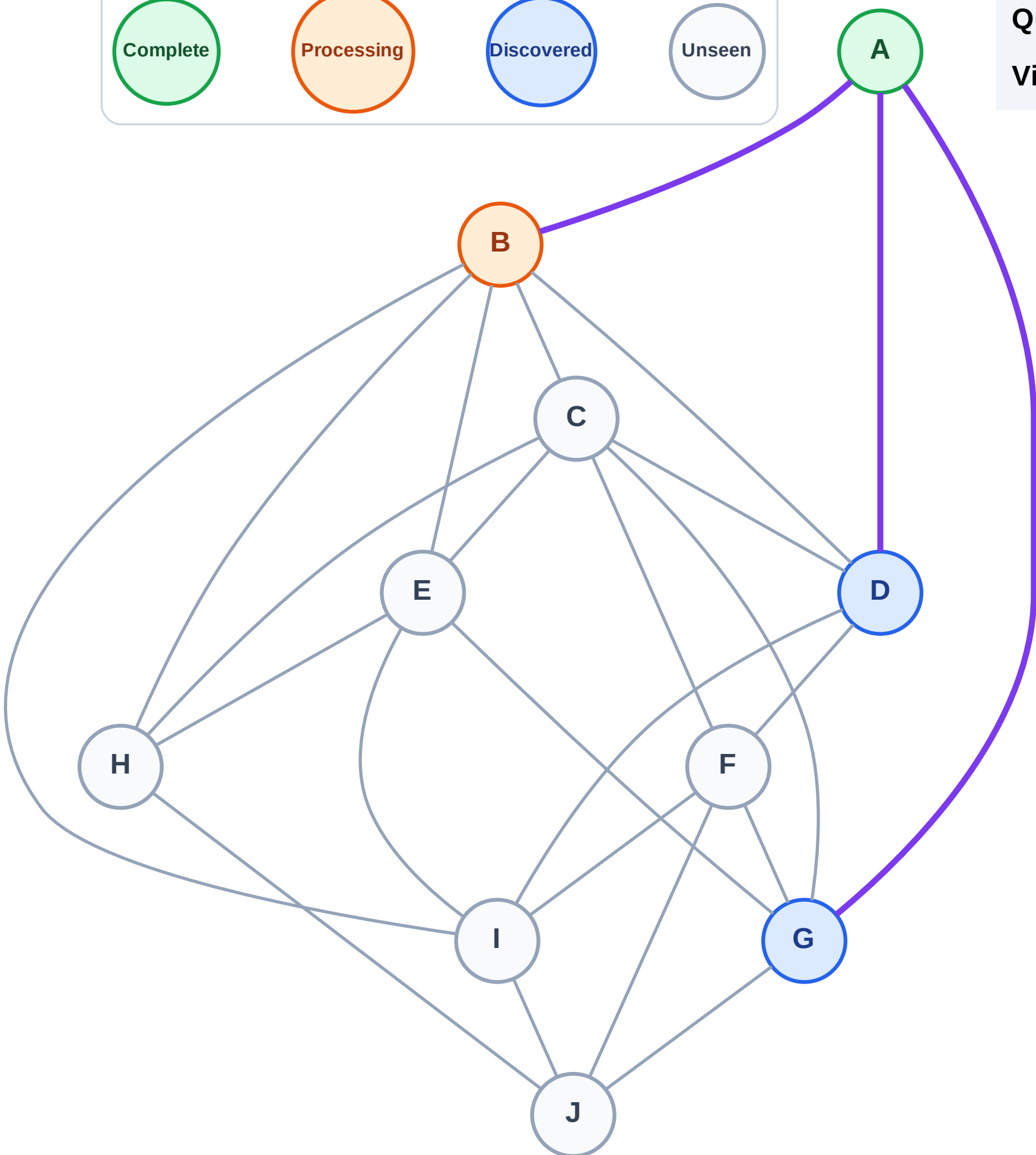
Step 4: Process node B

Legend



Queue: D → G

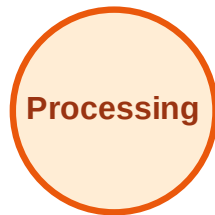
Visited: A



BFS Traversal

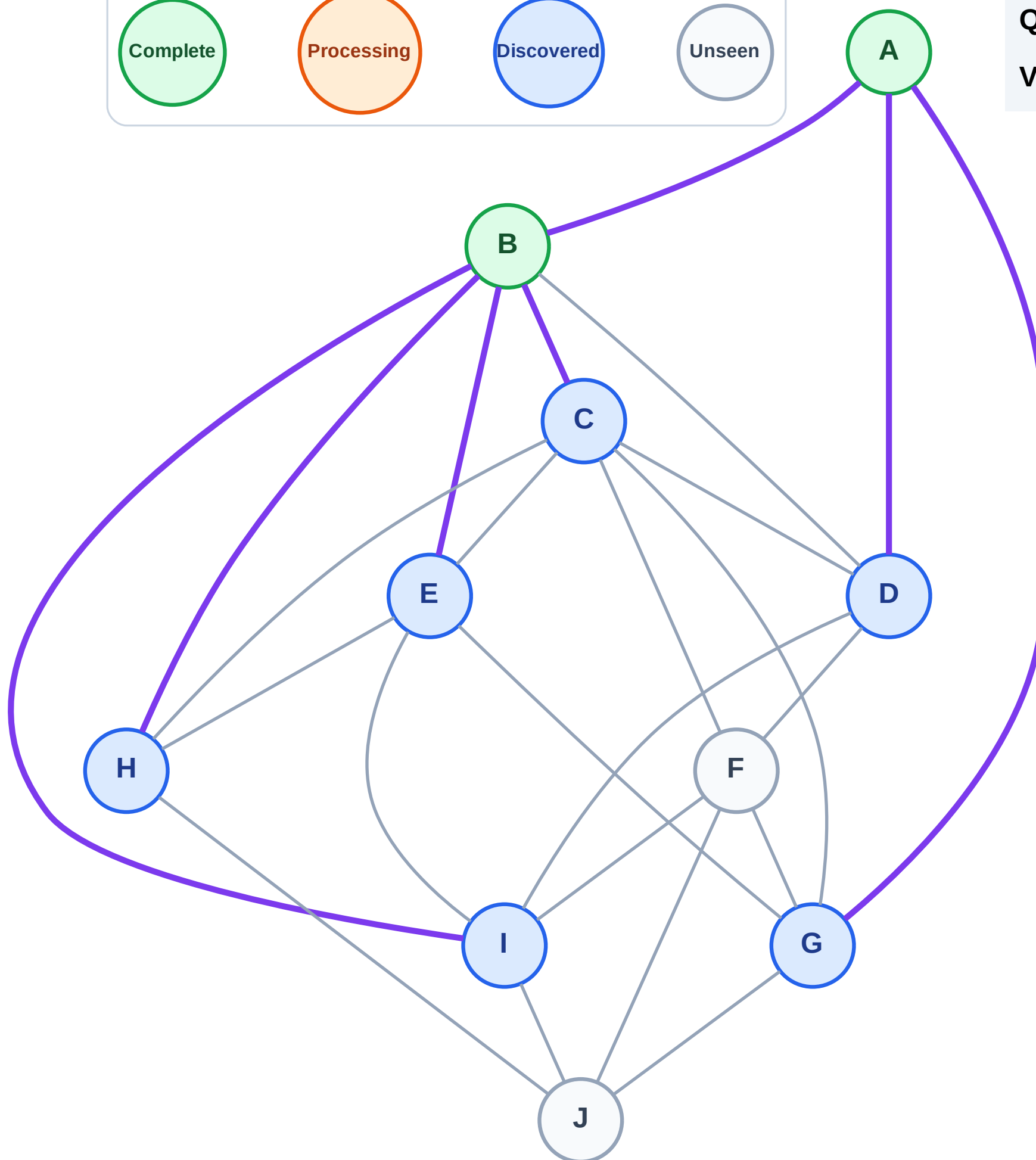
Step 5: Discover C, E, H, I from B

Legend



Queue: D → G → C → E → H → I

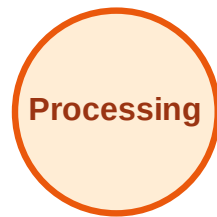
Visited: A, B



BFS Traversal

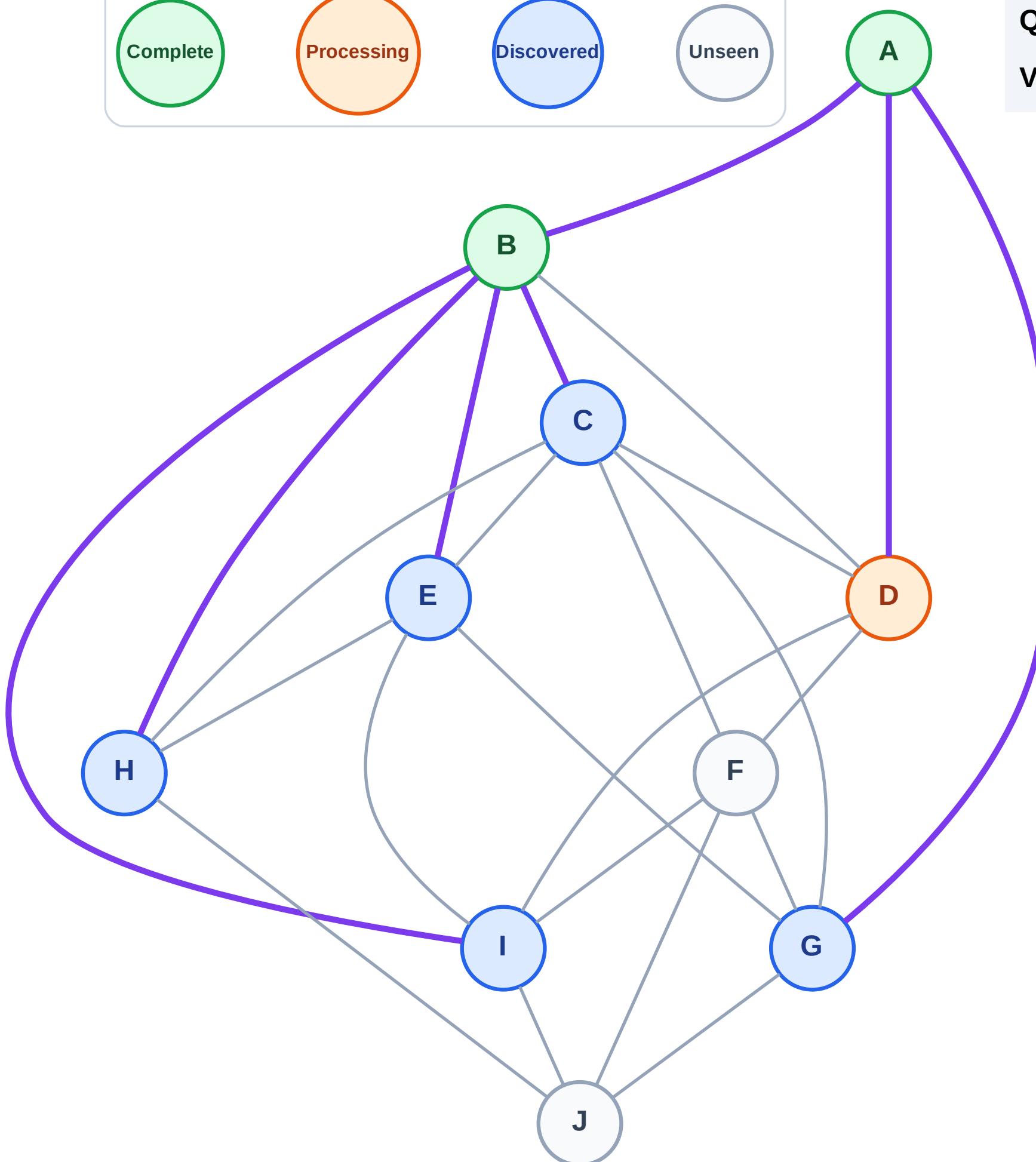
Step 6: Process node D

Legend



Queue: G → C → E → H → I

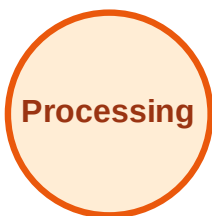
Visited: A, B



BFS Traversal

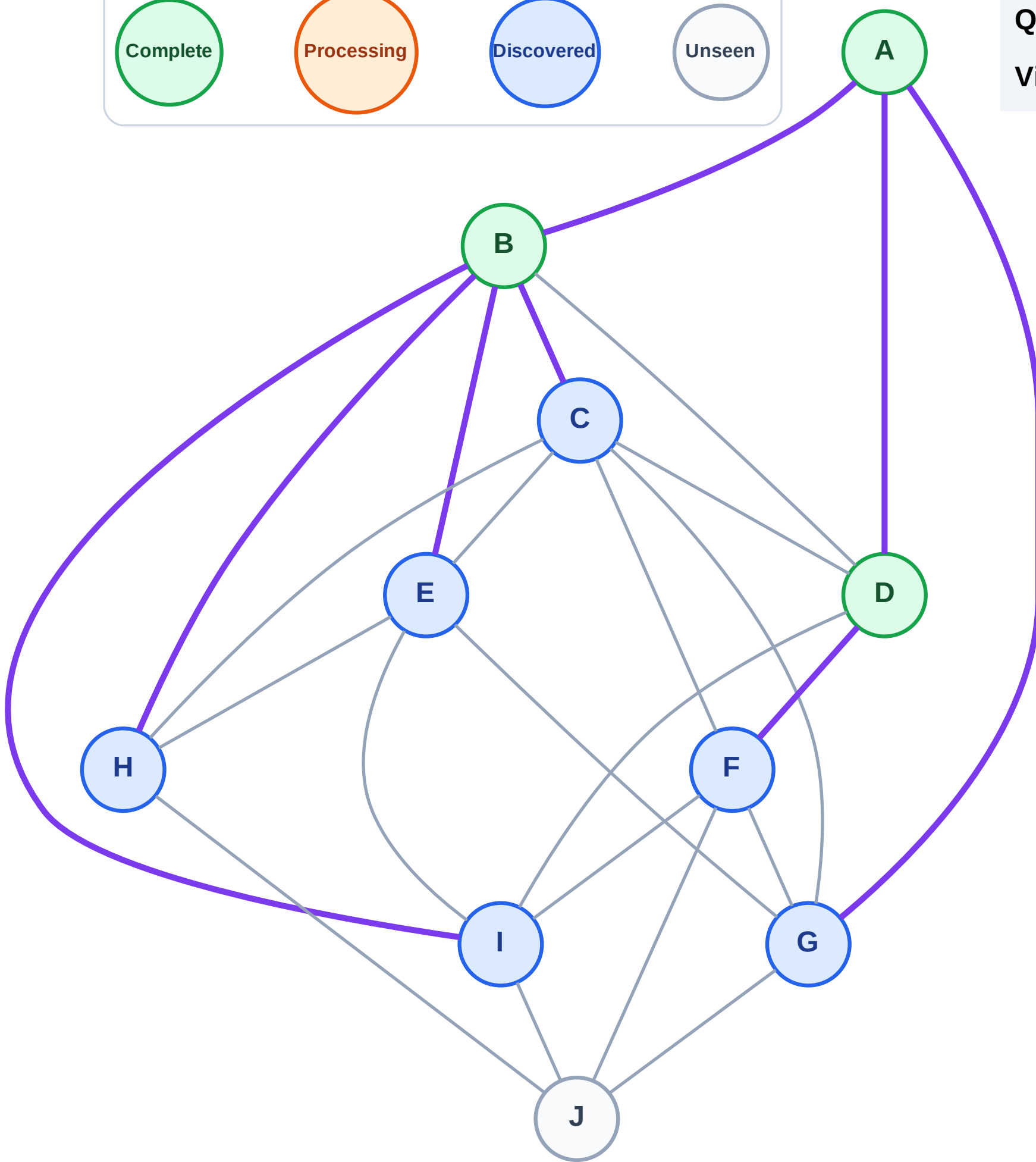
Step 7: Discover F from D

Legend



Queue: G → C → E → H → I → F

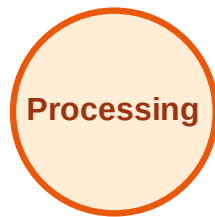
Visited: A, B, D



BFS Traversal

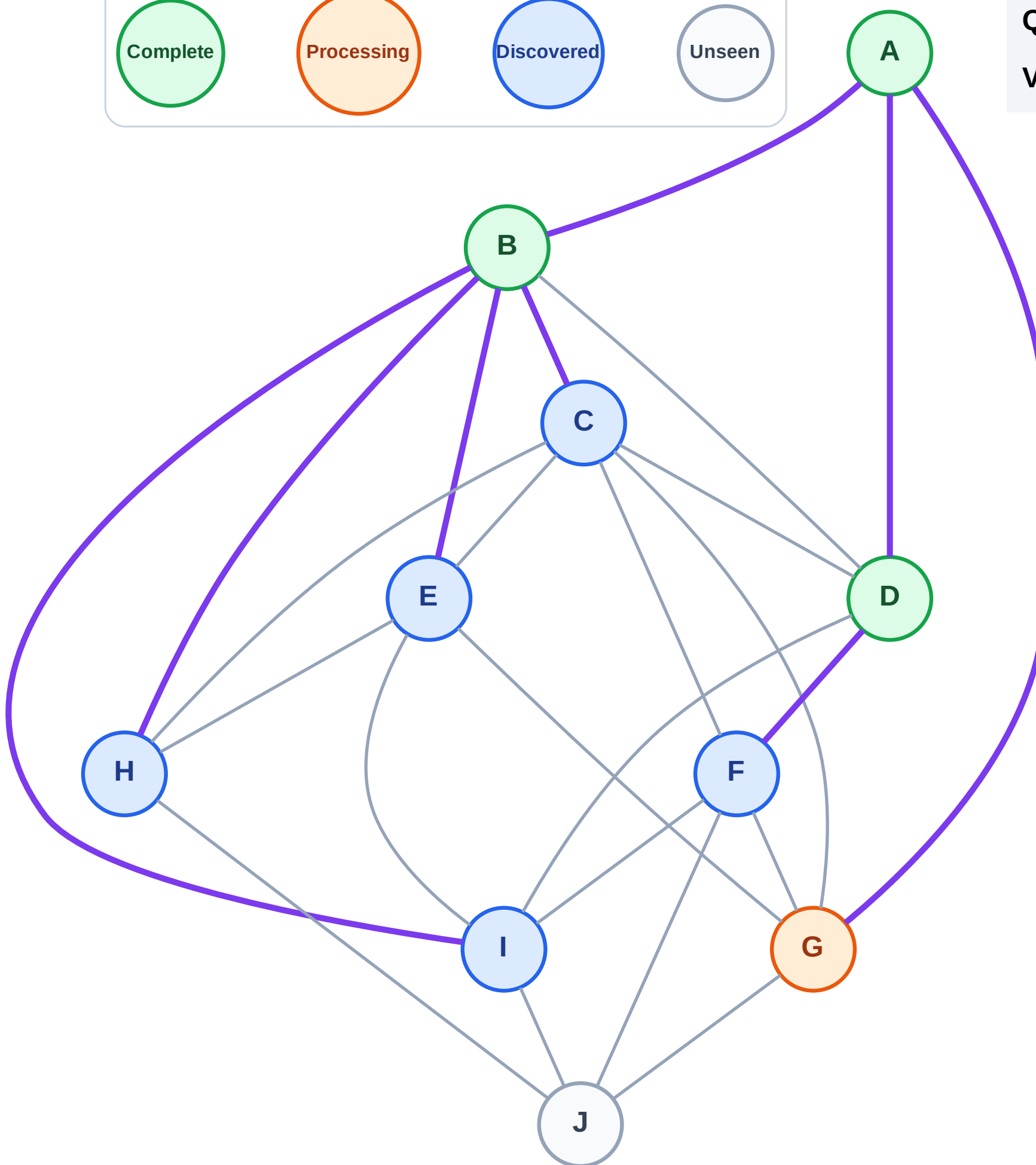
Step 8: Process node G

Legend



Queue: C → E → H → I → F

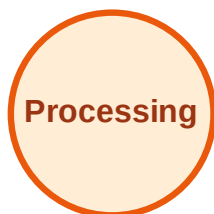
Visited: A, B, D



BFS Traversal

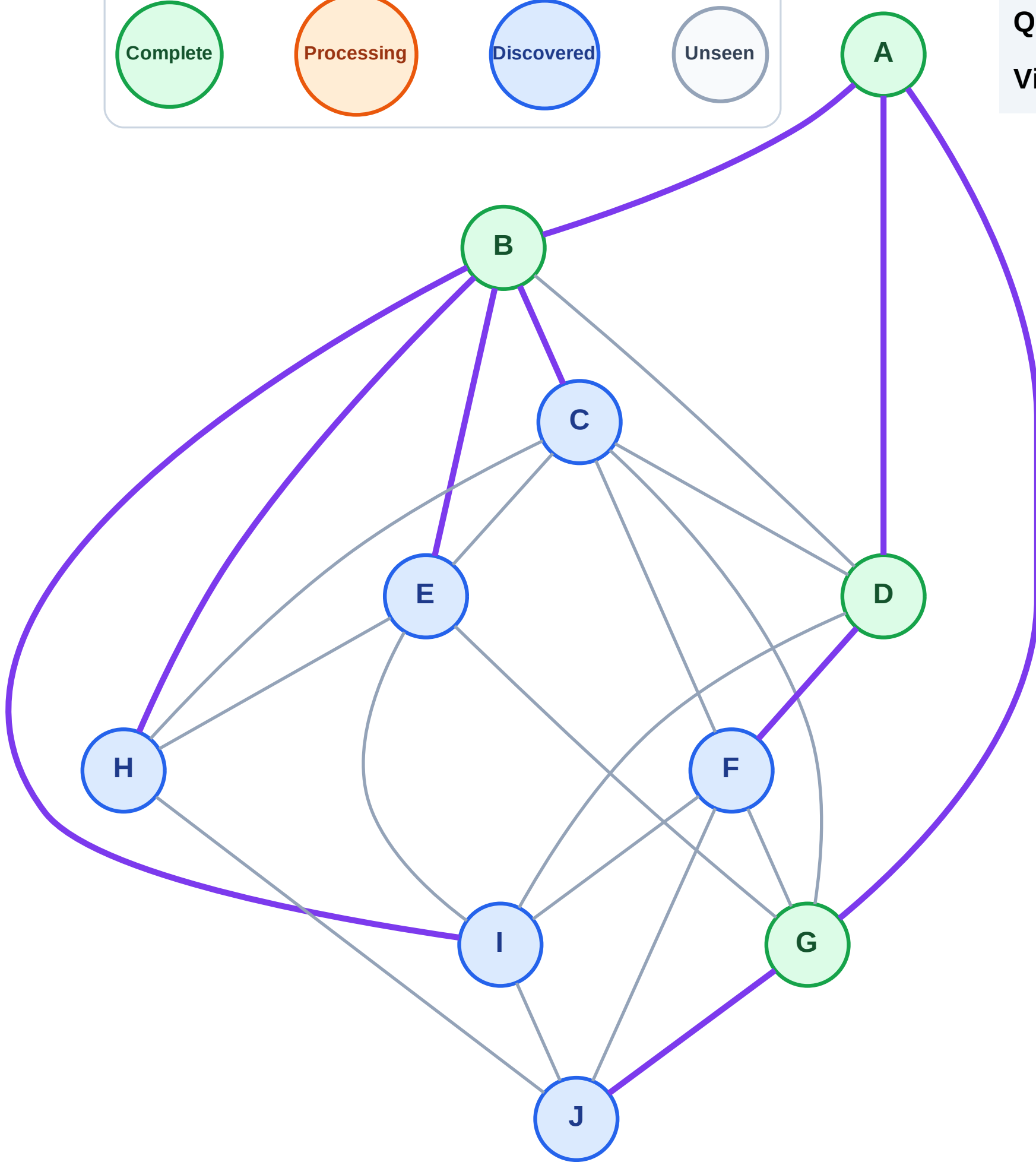
Step 9: Discover J from G

Legend



Queue: C → E → H → I → F → J

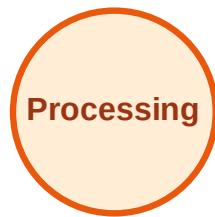
Visited: A, B, D, G



BFS Traversal

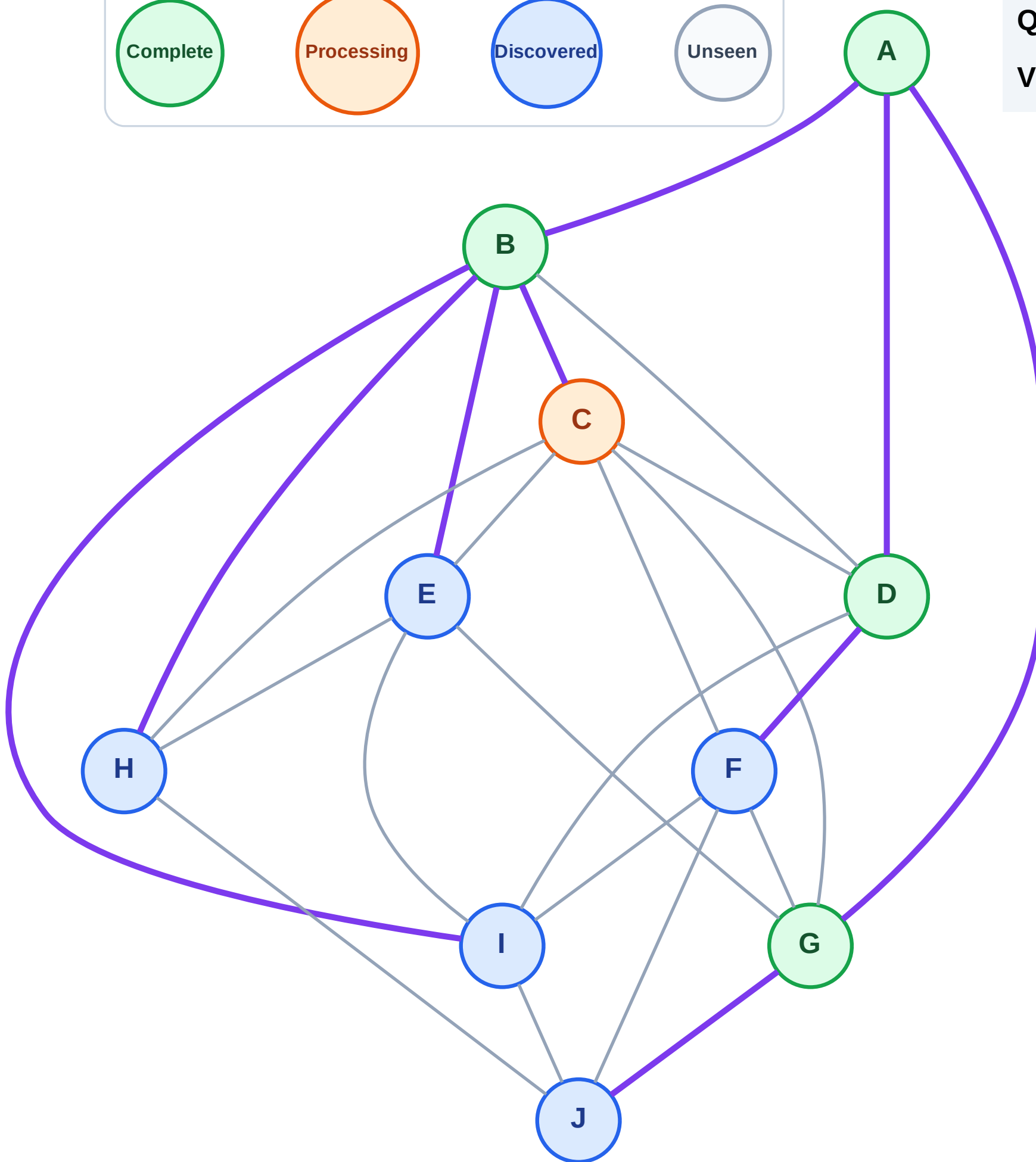
Step 10: Process node C

Legend



Queue: E → H → I → F → J

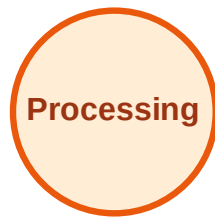
Visited: A, B, D, G



BFS Traversal

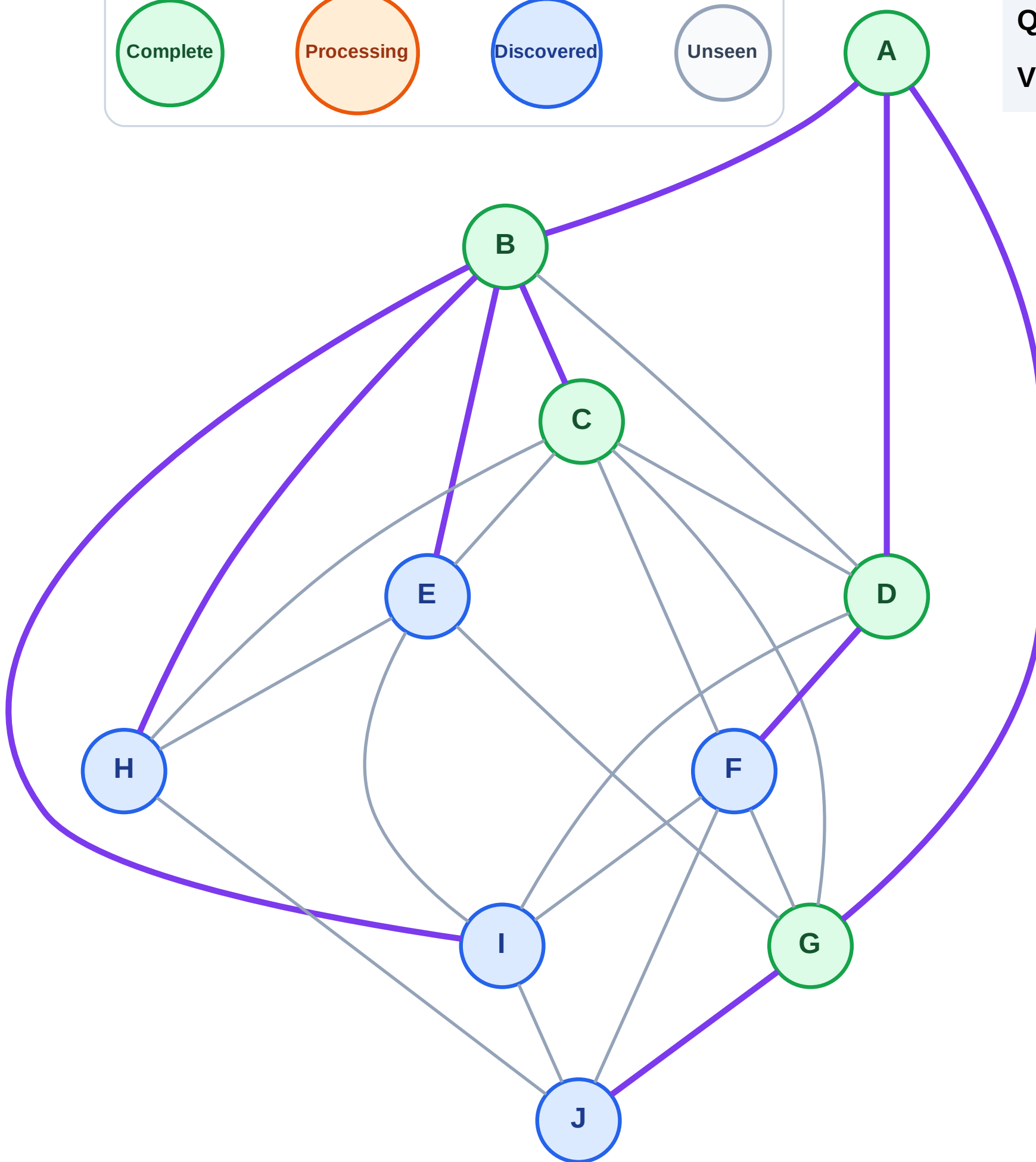
Step 11: No new neighbors from C

Legend



Queue: E → H → I → F → J

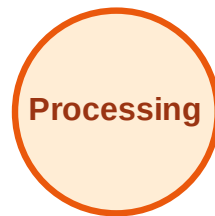
Visited: A, B, C, D, G



BFS Traversal

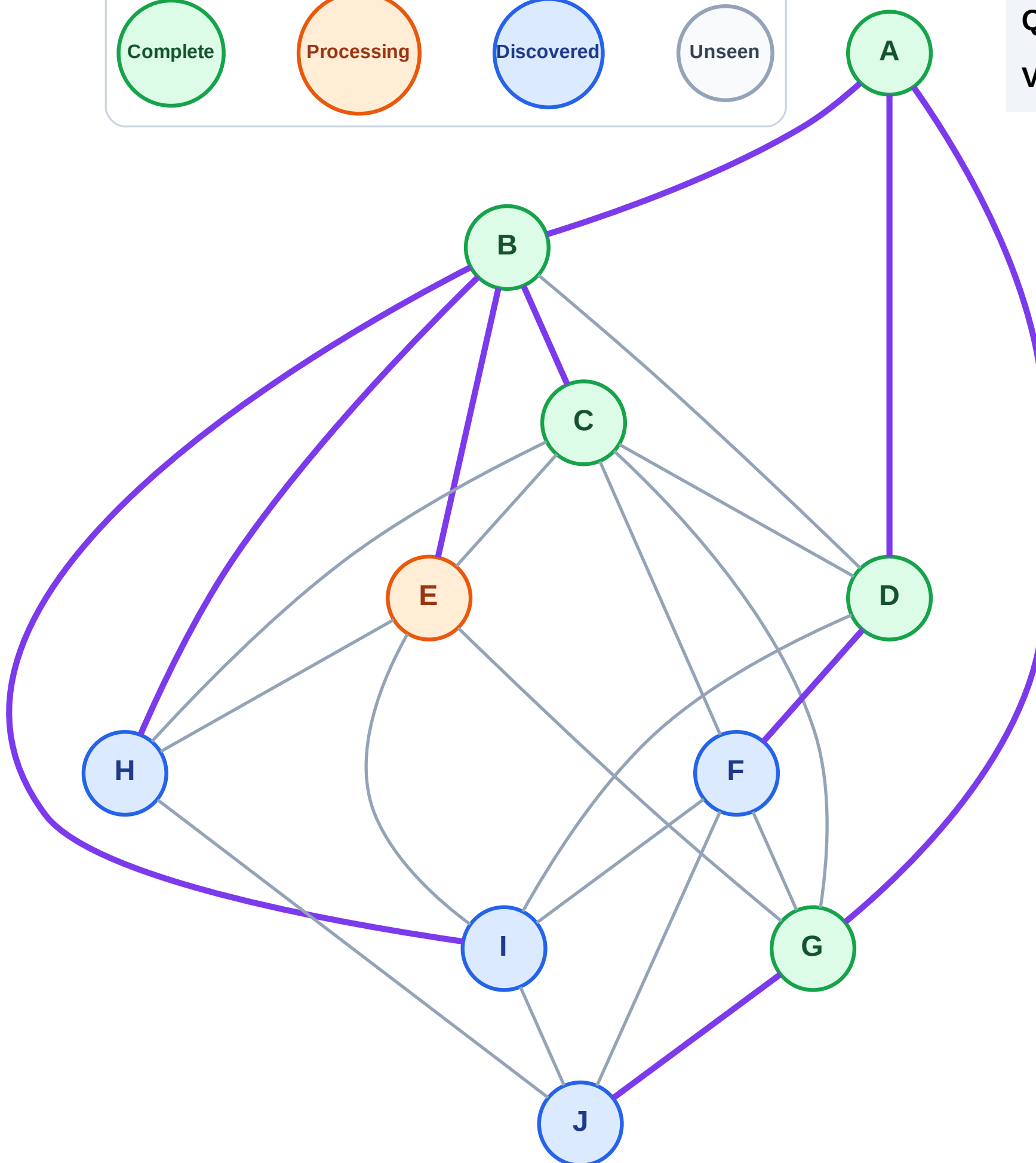
Step 12: Process node E

Legend



Queue: H → I → F → J

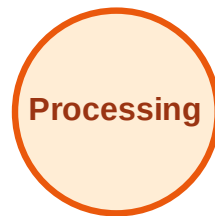
Visited: A, B, C, D, G



BFS Traversal

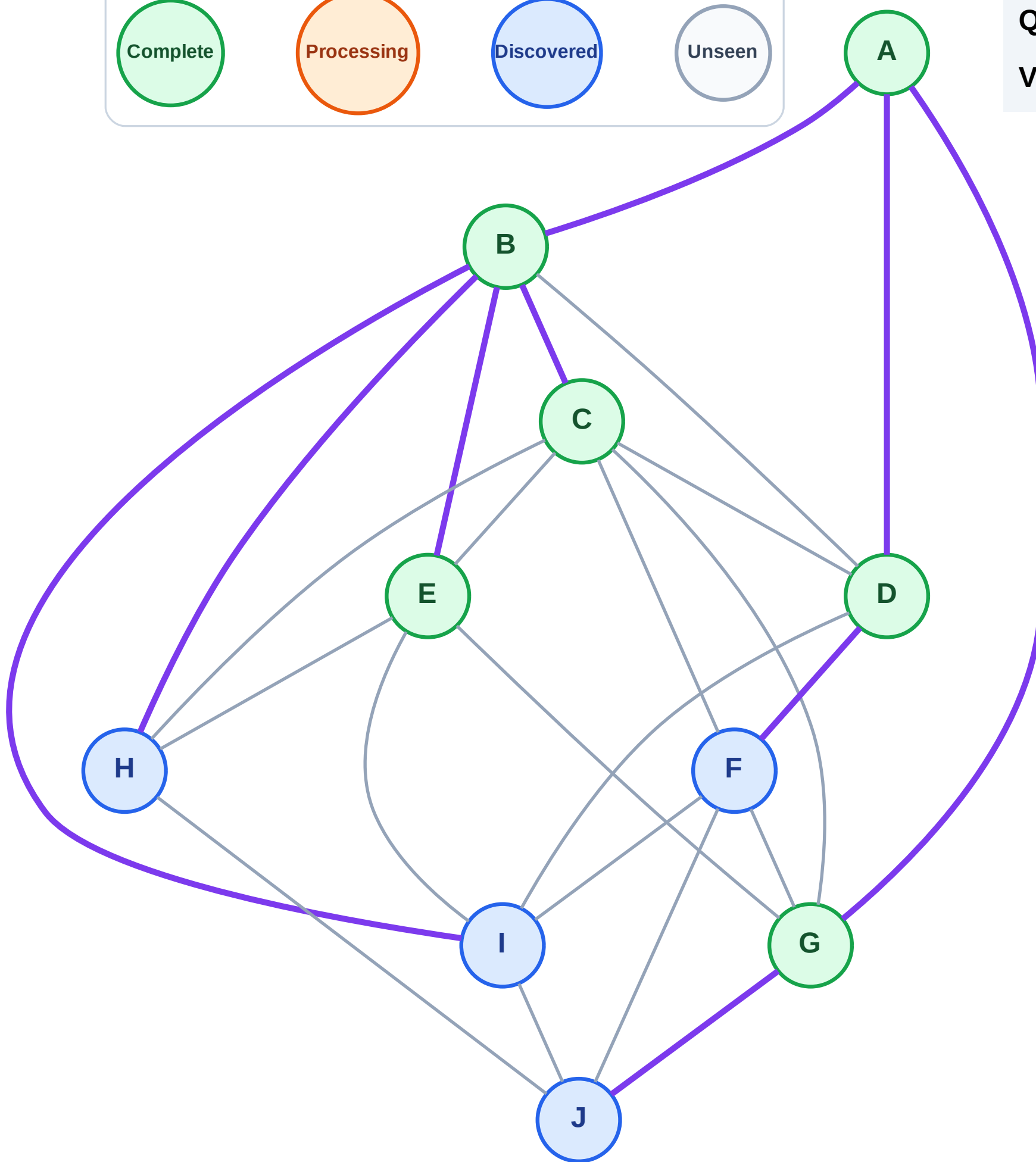
Step 13: No new neighbors from E

Legend



Queue: H → I → F → J

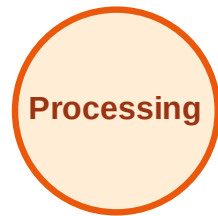
Visited: A, B, C, D, E, G



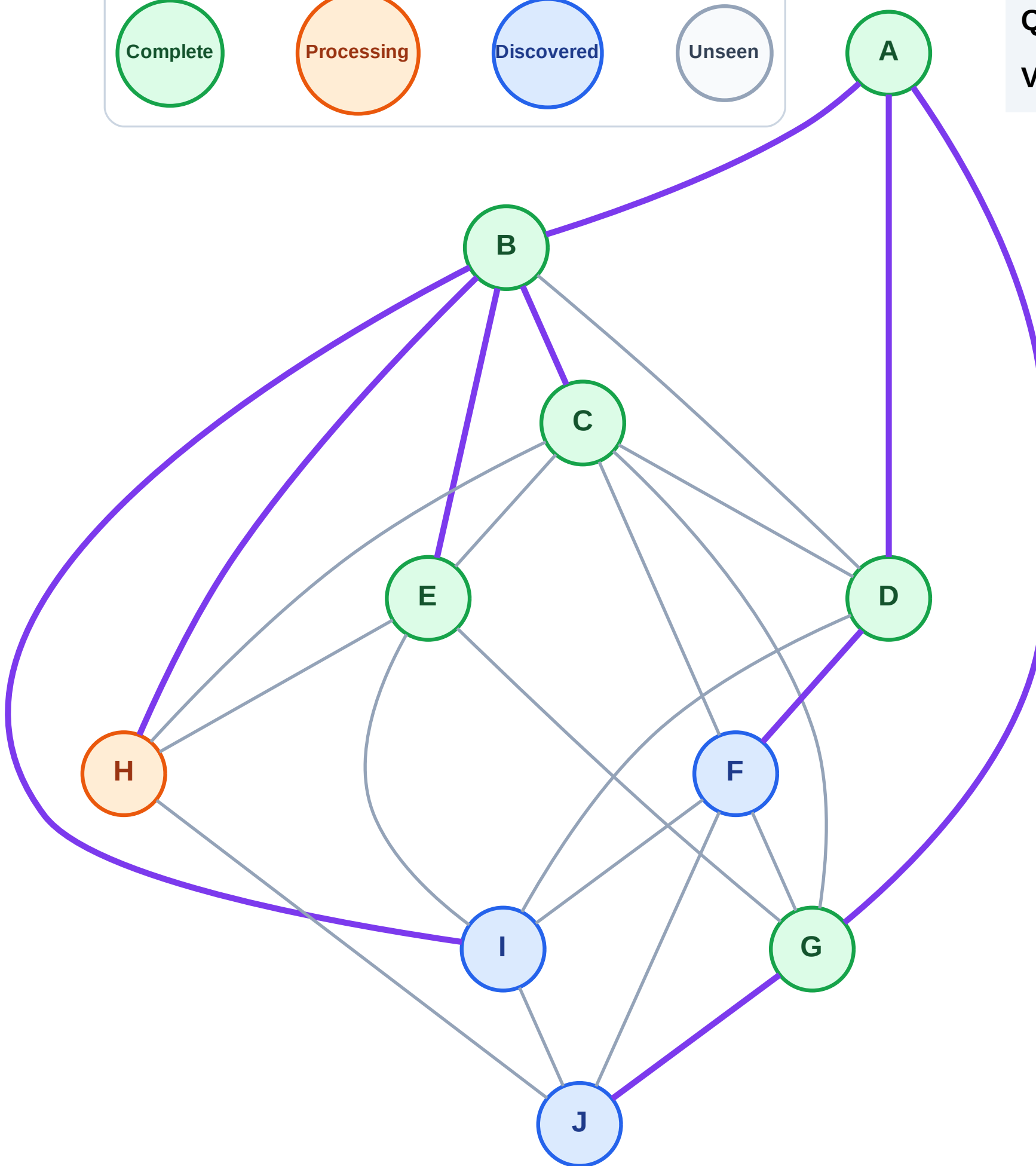
BFS Traversal

Step 14: Process node H

Legend



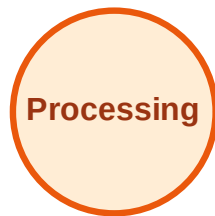
Queue: I → F → J
Visited: A, B, C, D, E, G



BFS Traversal

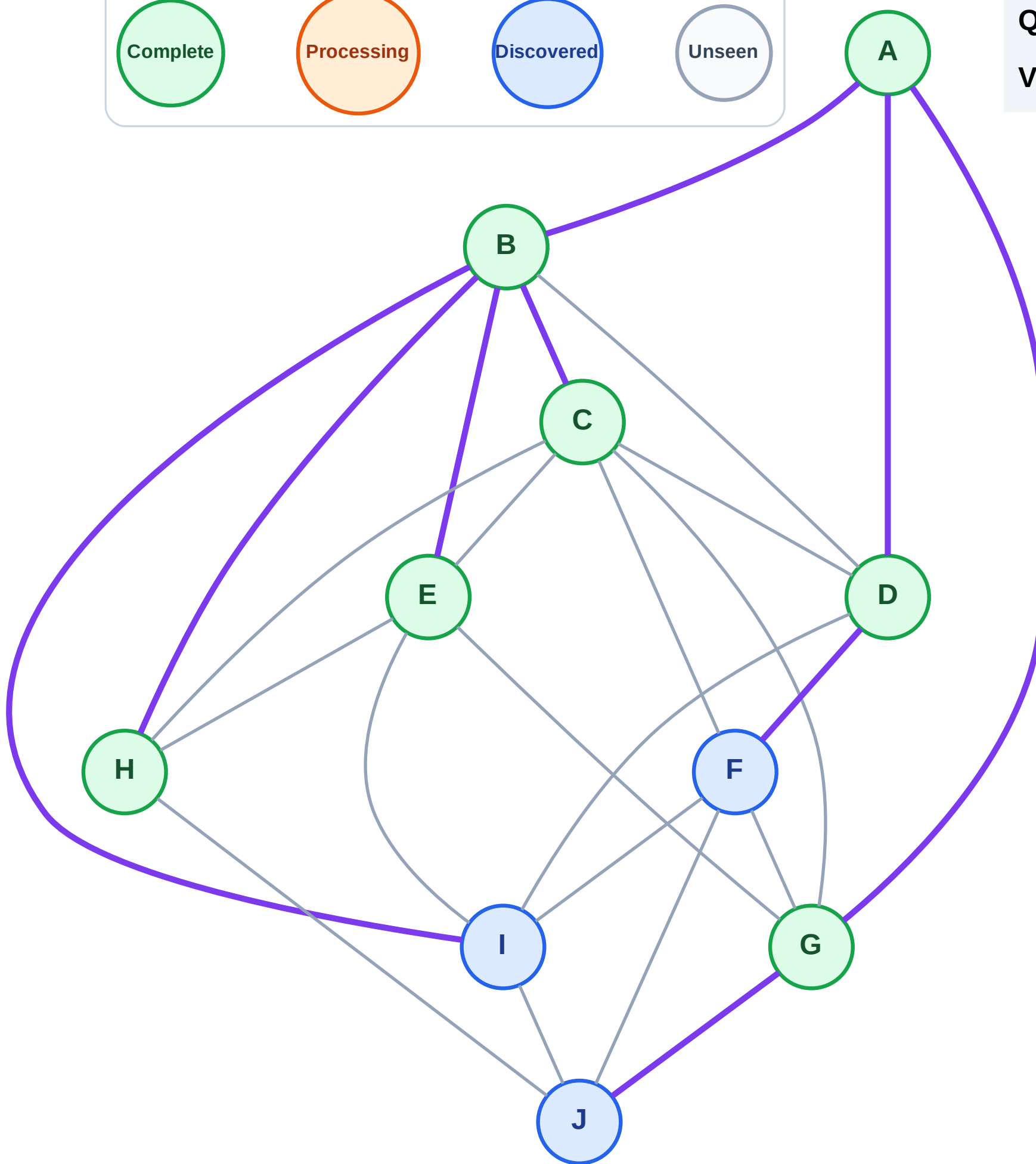
Step 15: No new neighbors from H

Legend



Queue: I → F → J

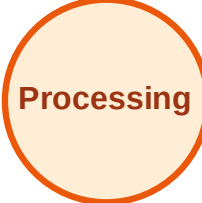
Visited: A, B, C, D, E, G, H



BFS Traversal

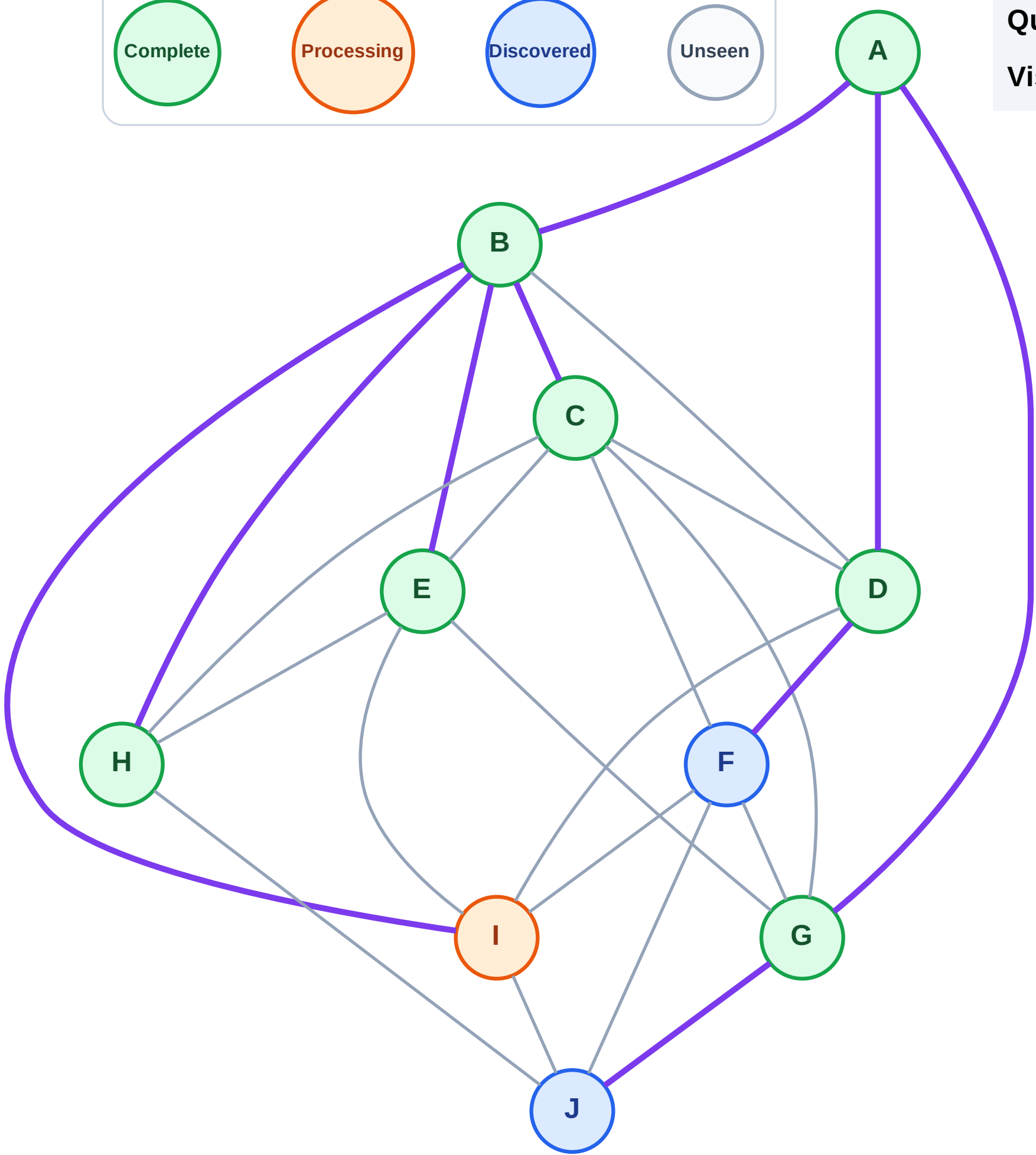
Step 16: Process node I

Legend



Queue: F → J

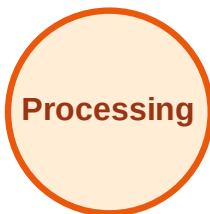
Visited: A, B, C, D, E, G, H



BFS Traversal

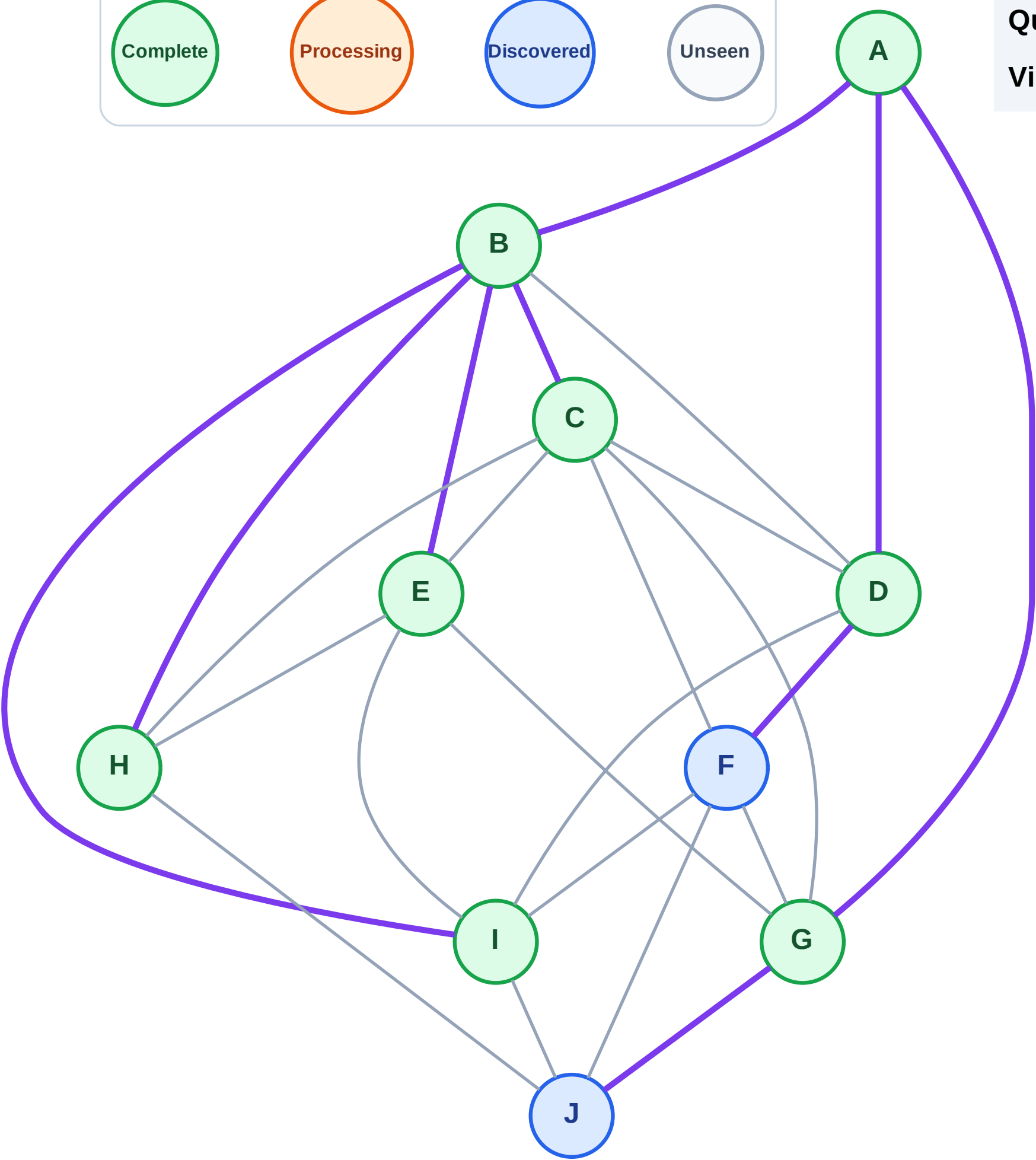
Step 17: No new neighbors from I

Legend



Queue: F → J

Visited: A, B, C, D, E, G, H, I



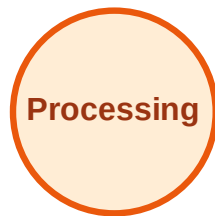
BFS Traversal

Step 18: Process node F

Legend



Complete



Processing



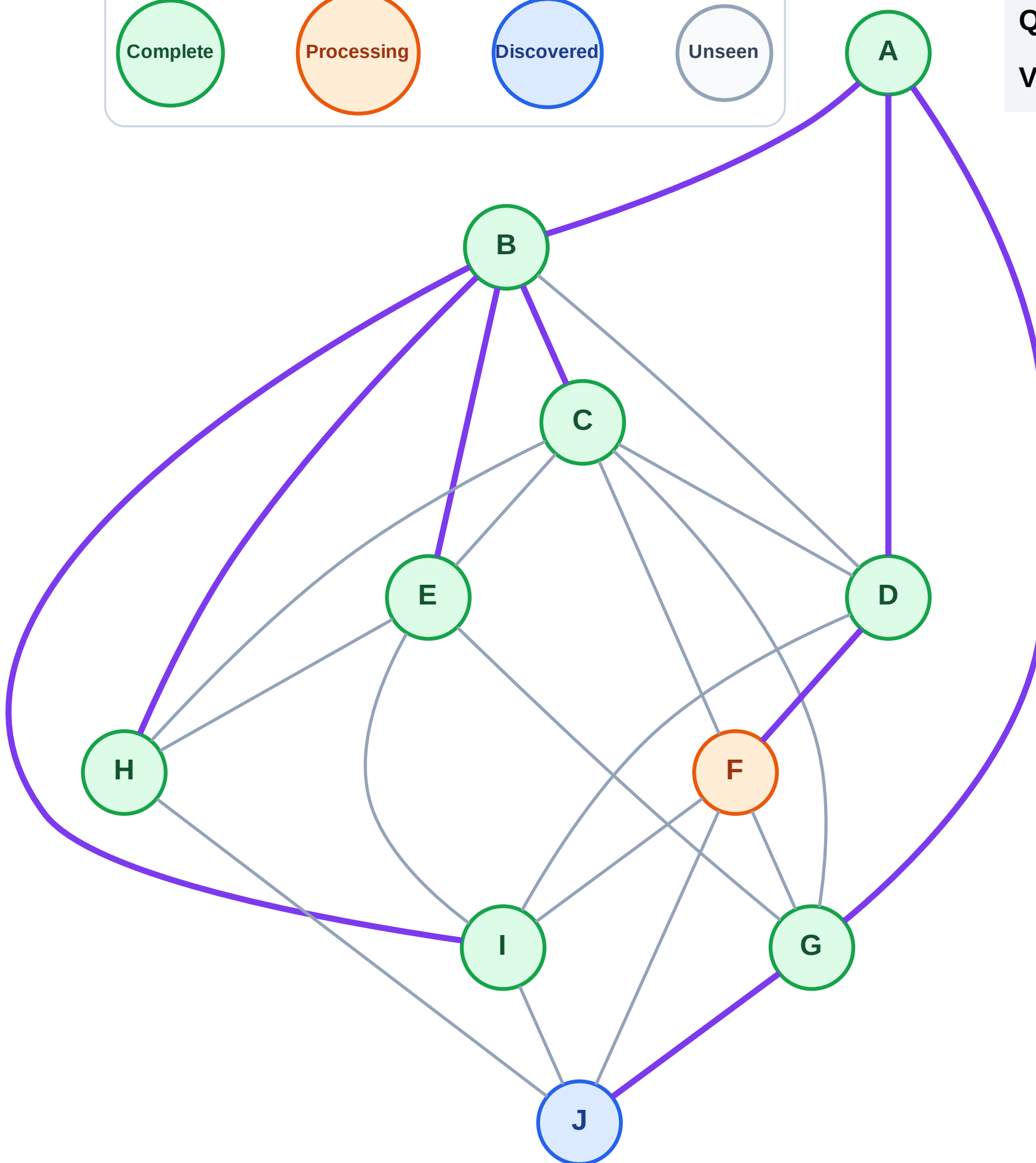
Discovered



Unseen

Queue: J

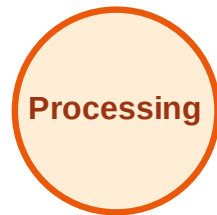
Visited: A, B, C, D, E, G, H, I



BFS Traversal

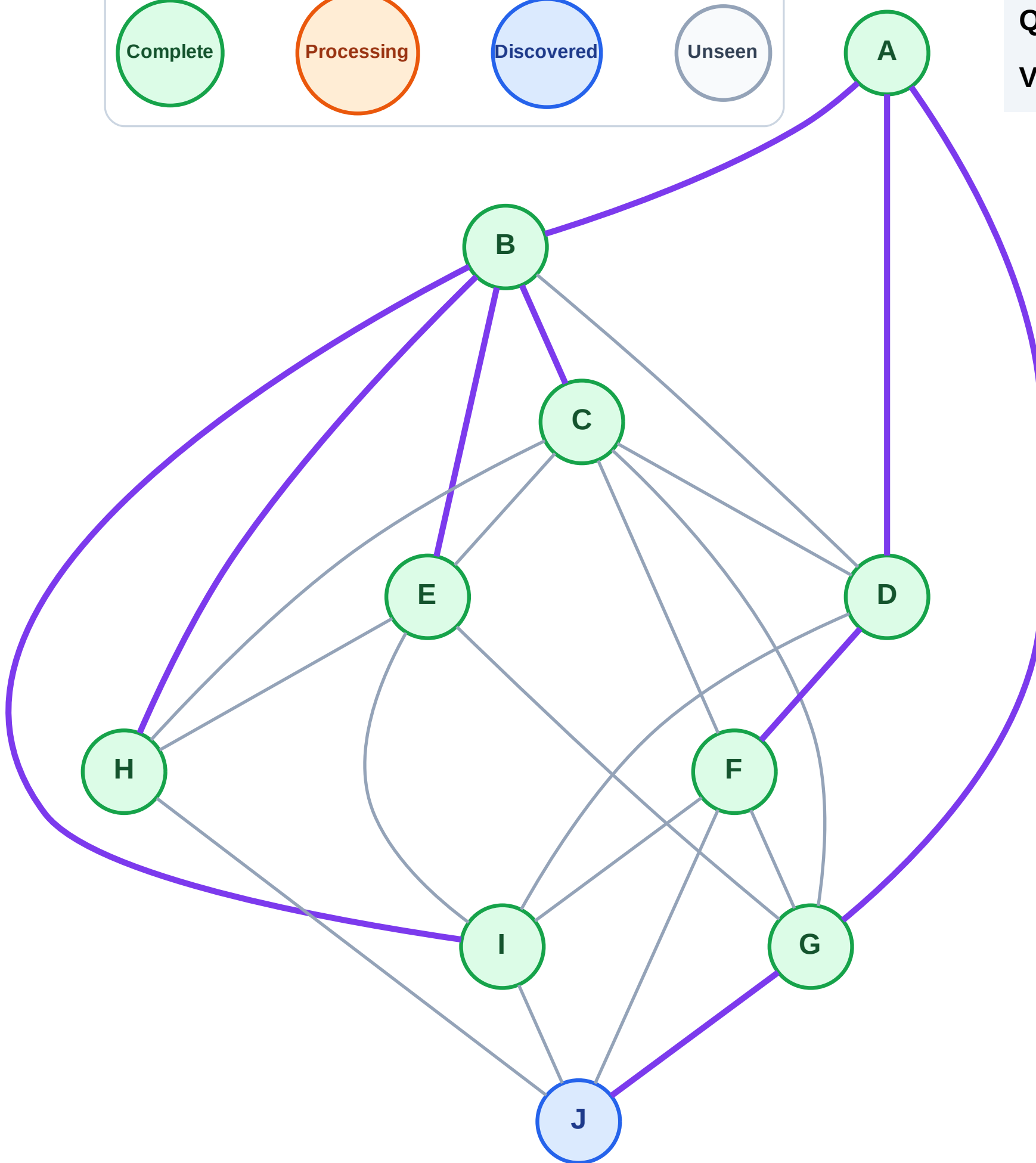
Step 19: No new neighbors from F

Legend



Queue: J

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

Step 20: Process node J

Legend

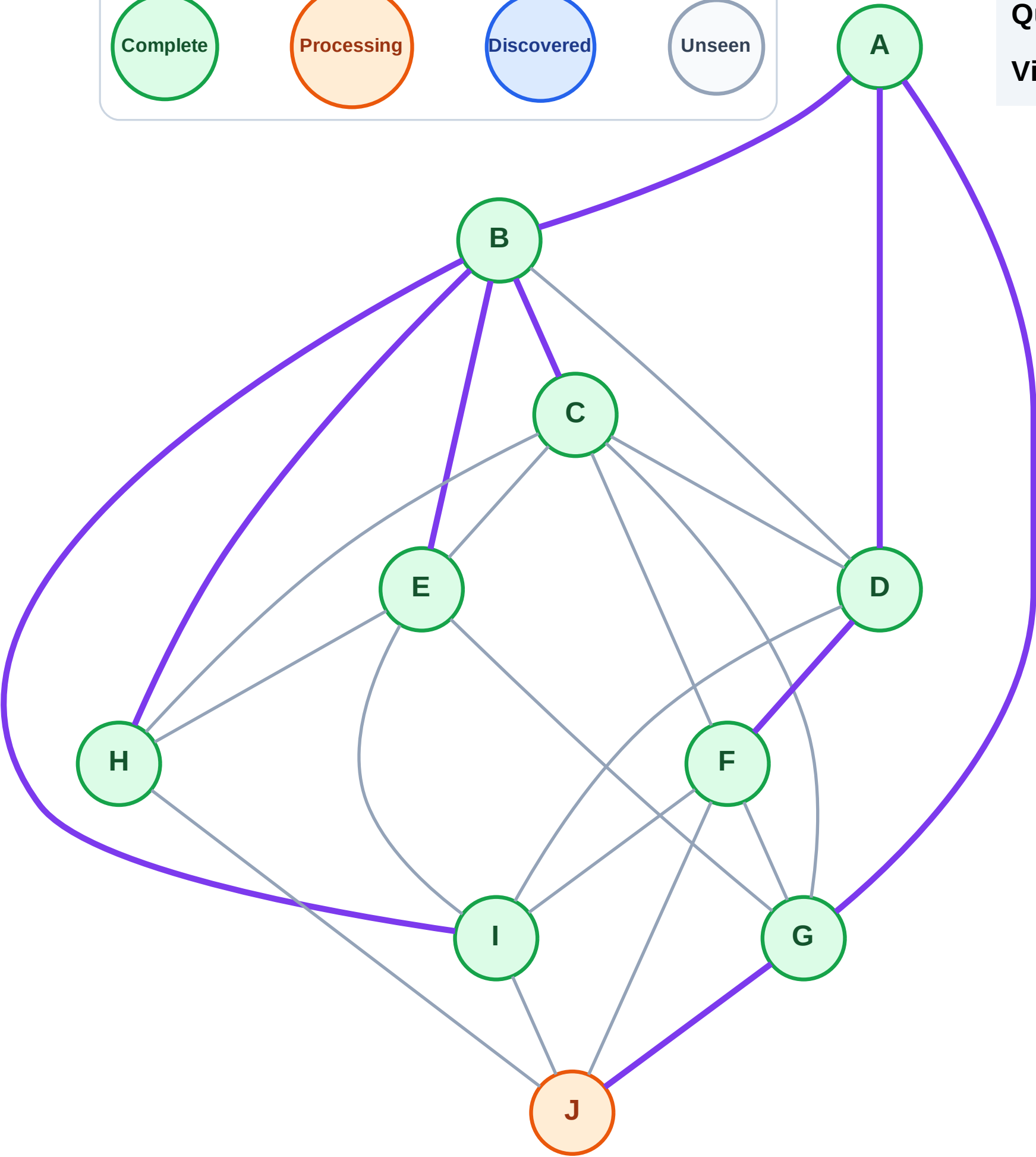
Complete

Processing

Discovered

Unseen

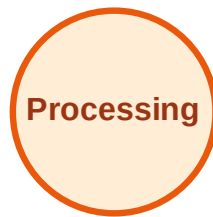
Queue: (empty)
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

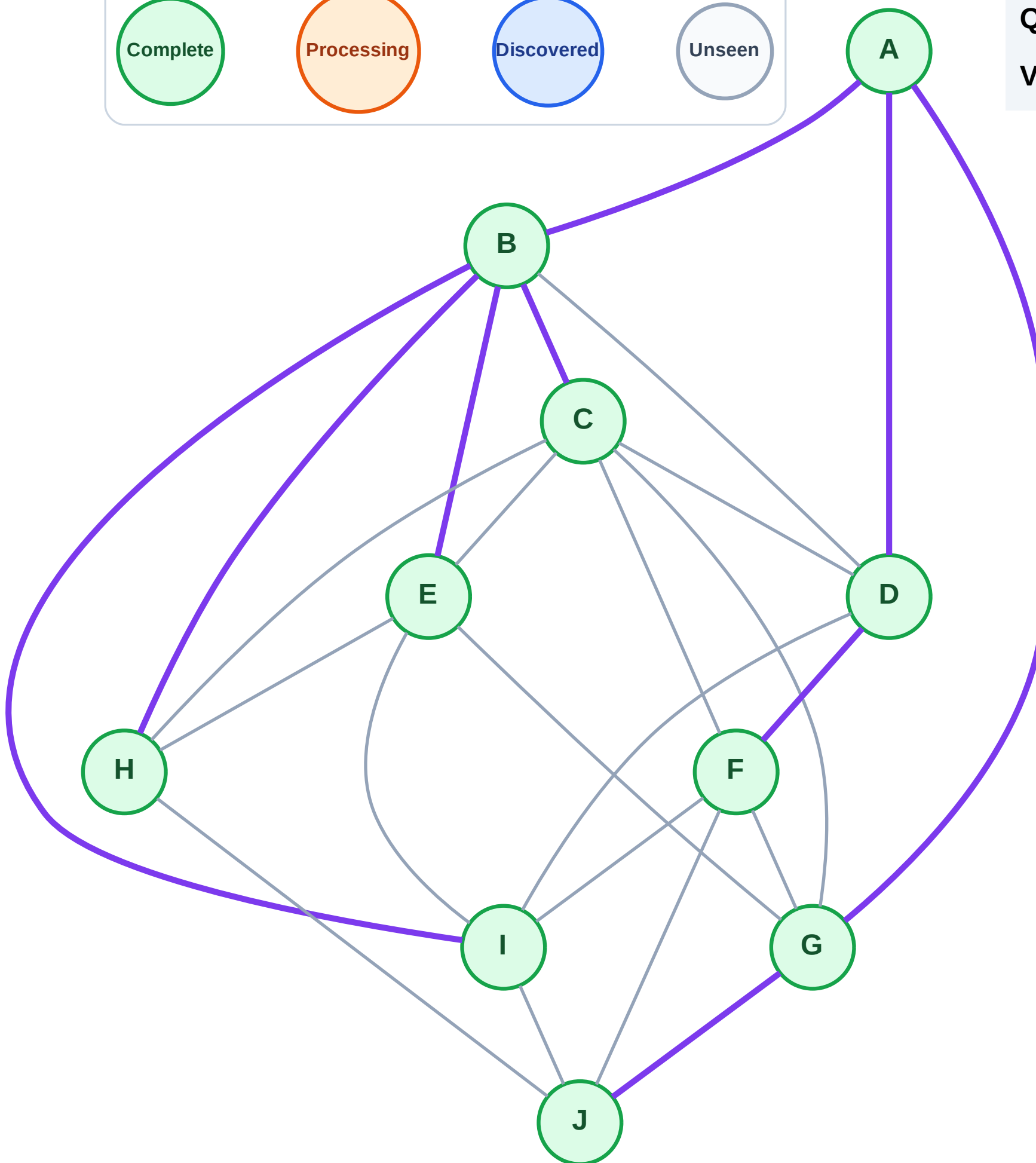
Step 21: No new neighbors from J

Legend



Queue: (empty)

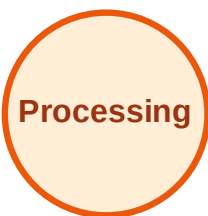
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

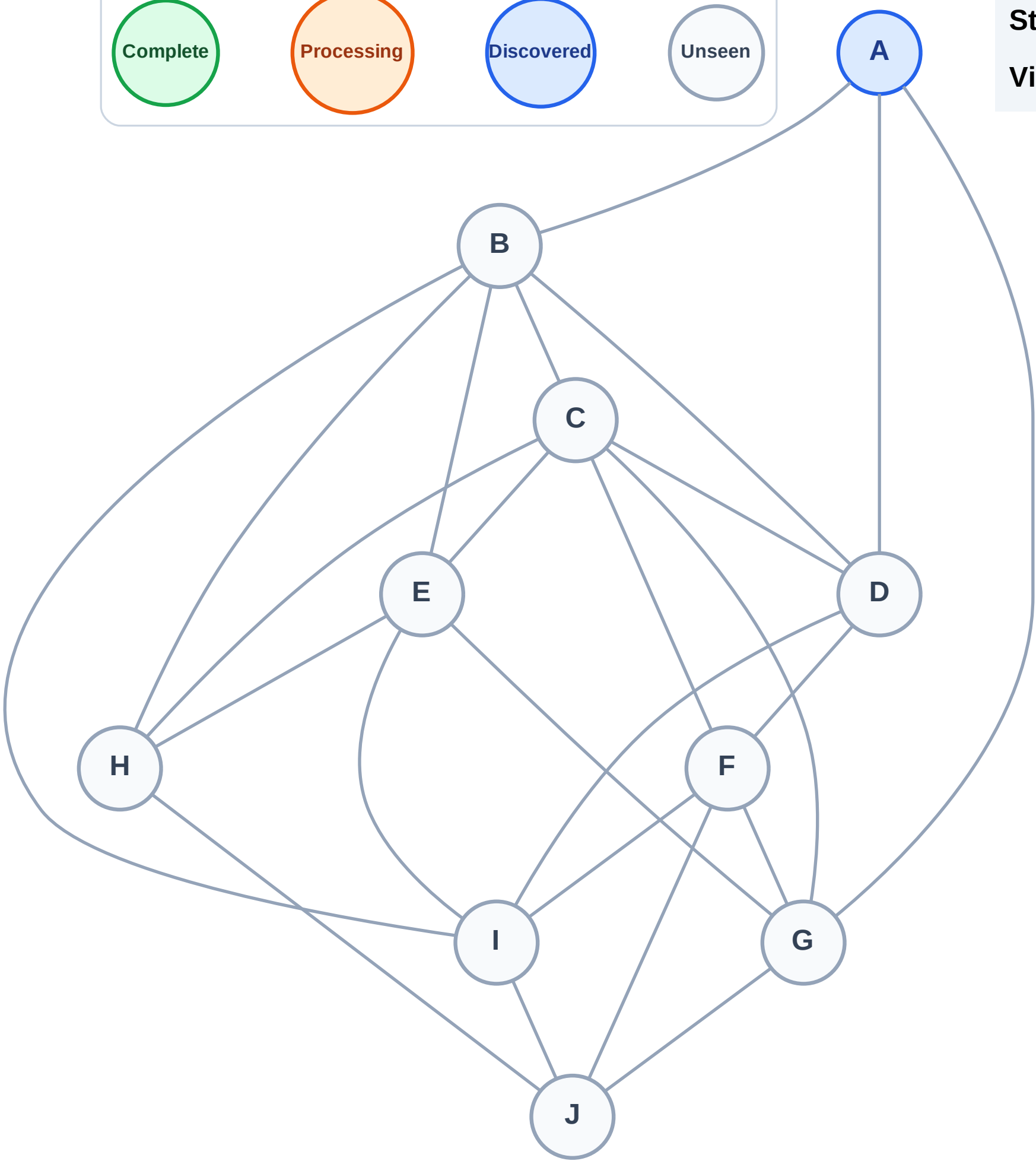
Step 1: Start at A

Legend



Stack (top at right): A

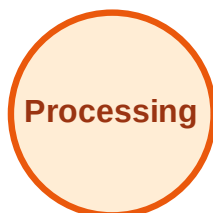
Visited: (none)



DFS Traversal

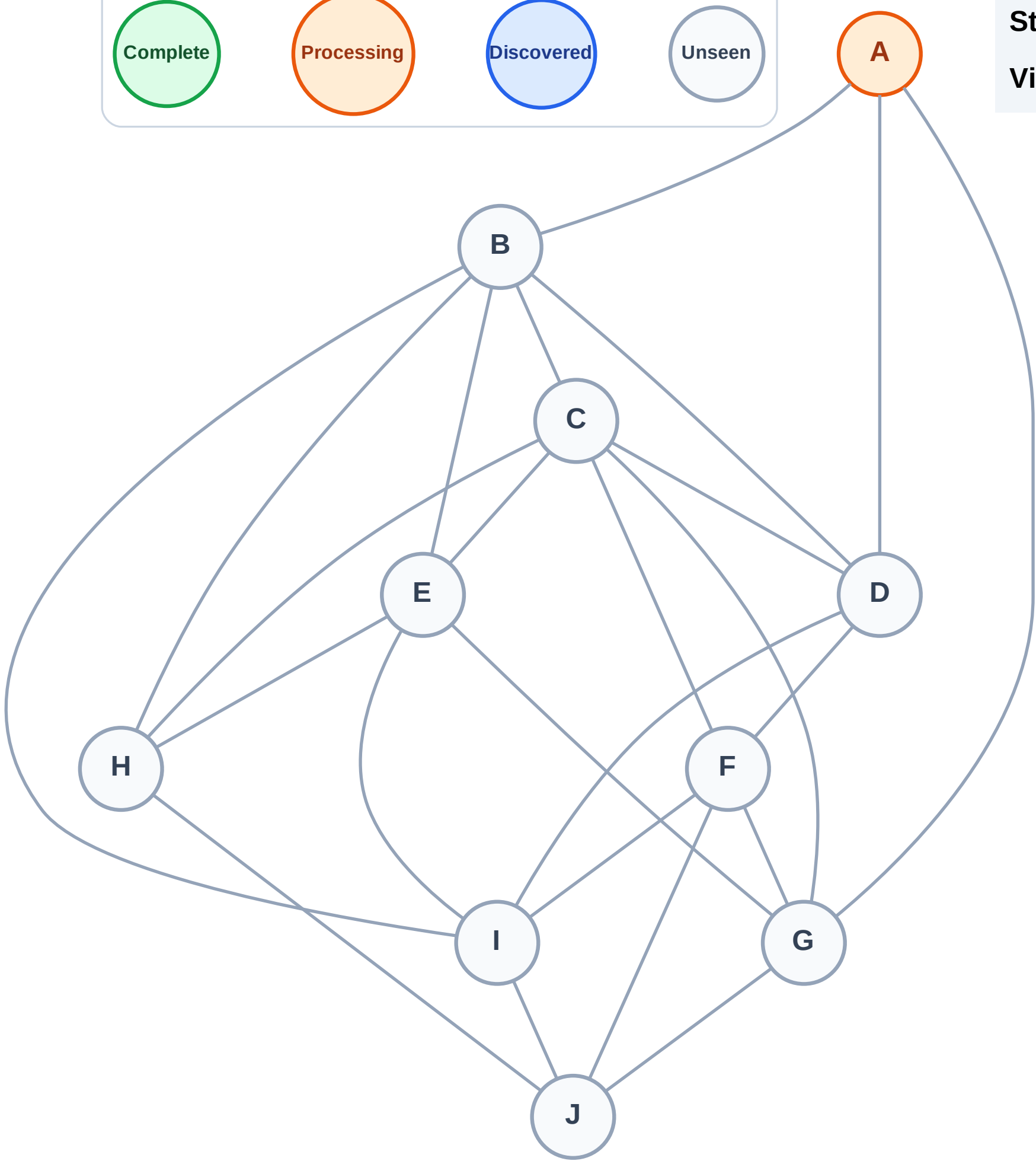
Step 2: Process node A

Legend



Stack (top at right): (empty)

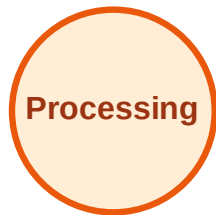
Visited: (none)



DFS Traversal

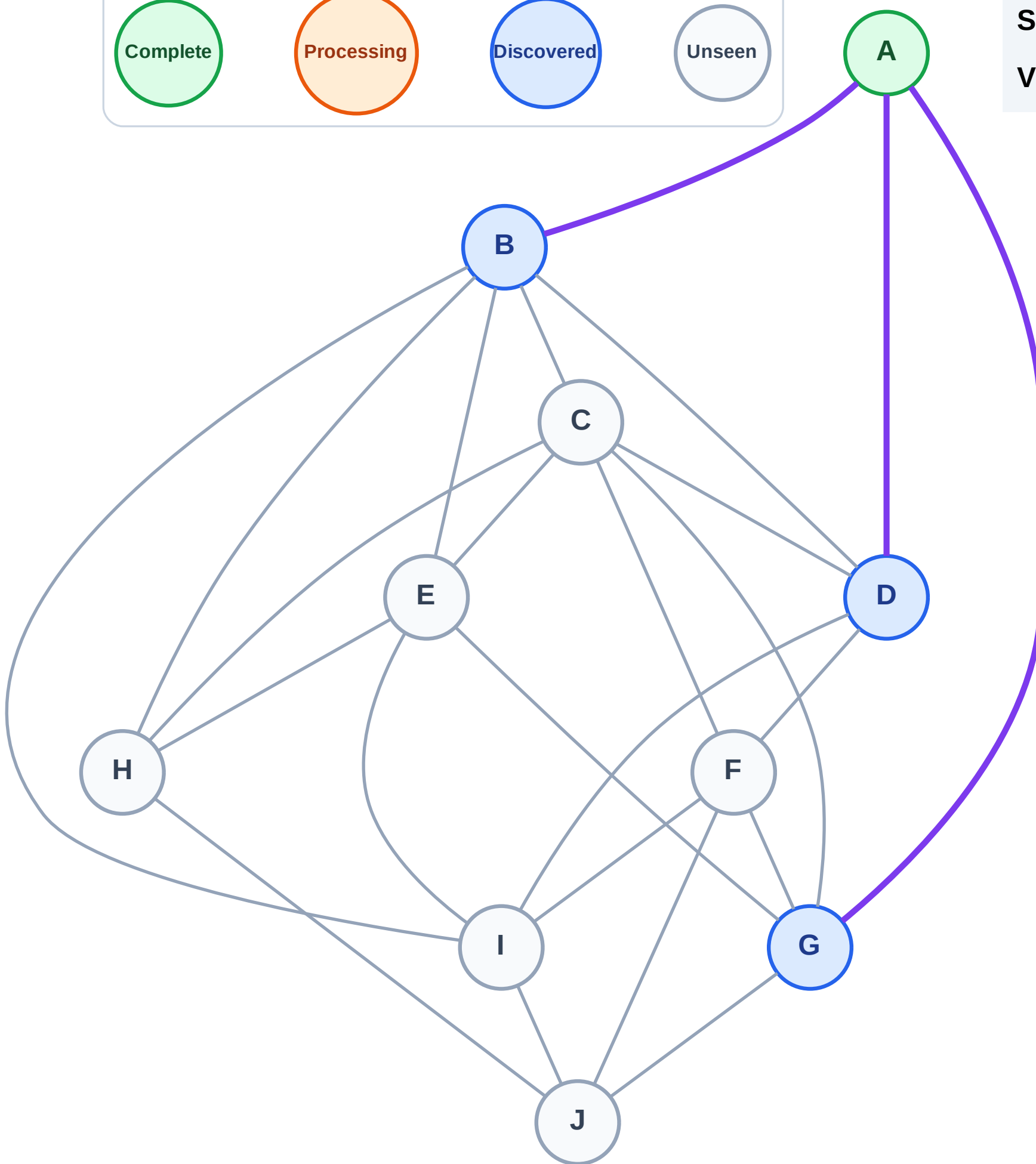
Step 3: Discover G, D, B from A

Legend



Stack (top at right): G | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

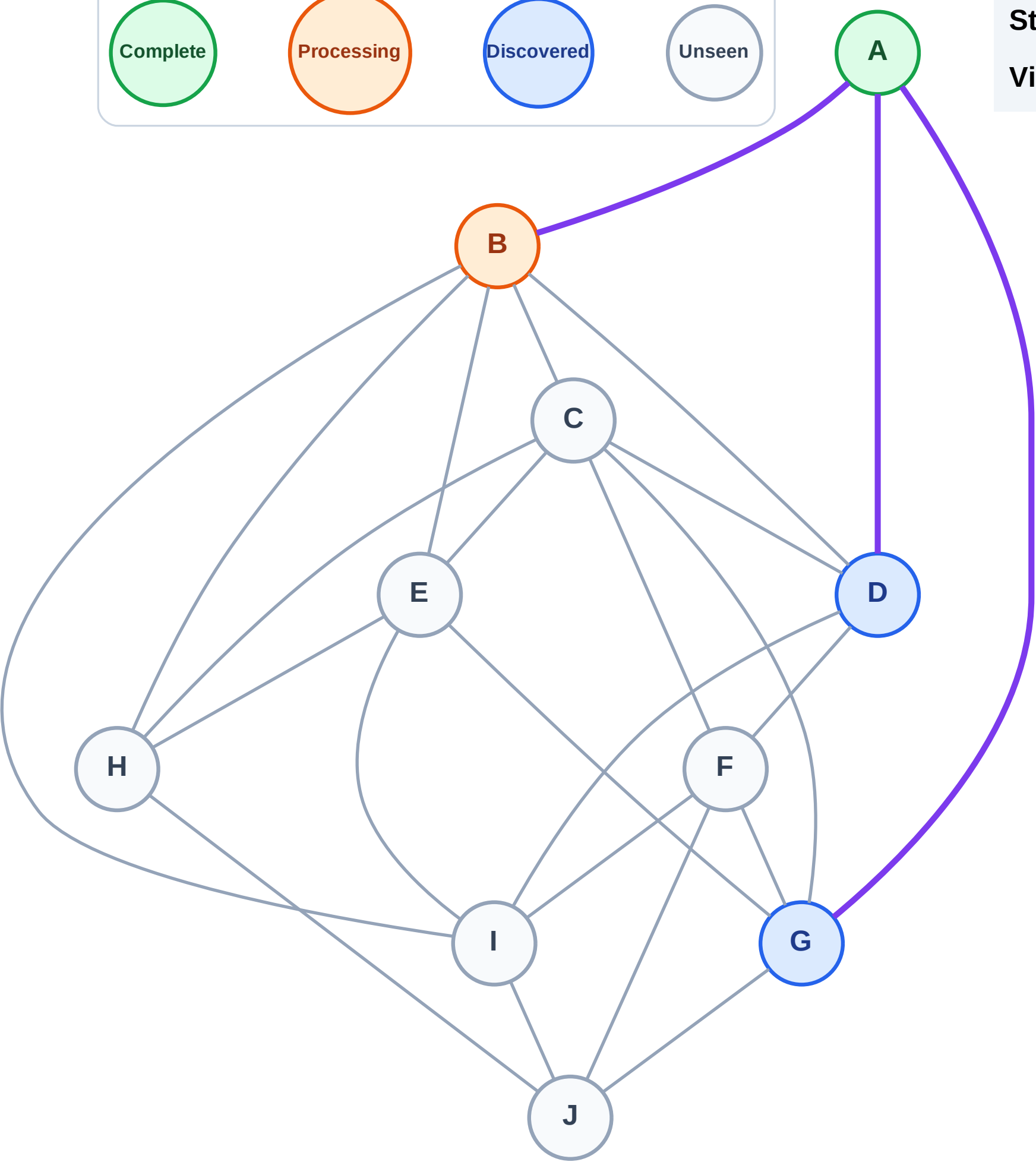
Processing

Discovered

Unseen

Stack (top at right): G | D

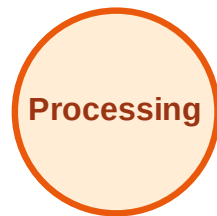
Visited: A



DFS Traversal

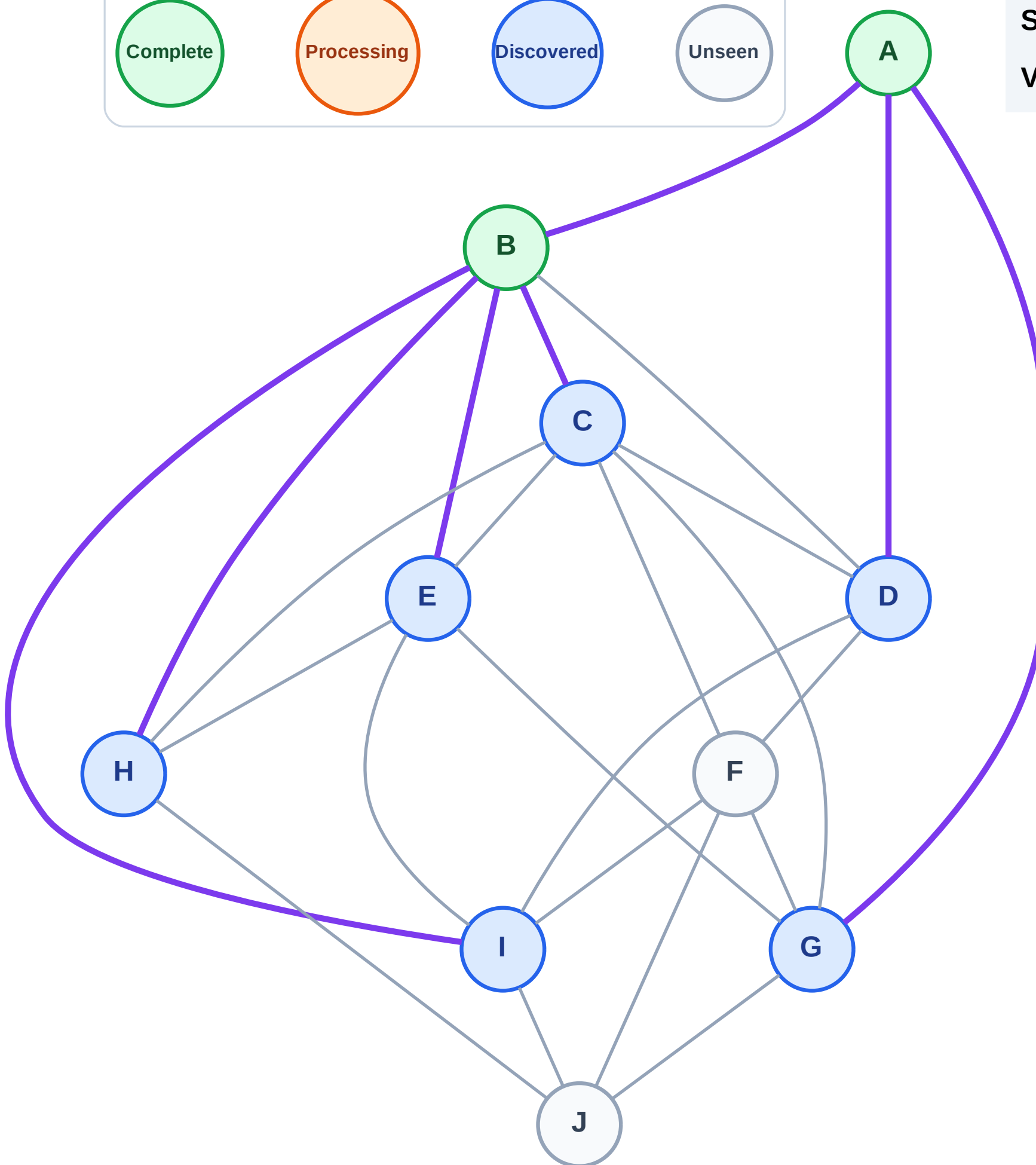
Step 5: Discover I, H, E, C from B

Legend



Stack (top at right): G | D | I | H | E | C

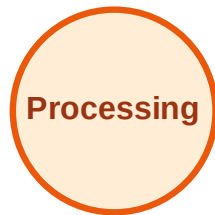
Visited: A, B



DFS Traversal

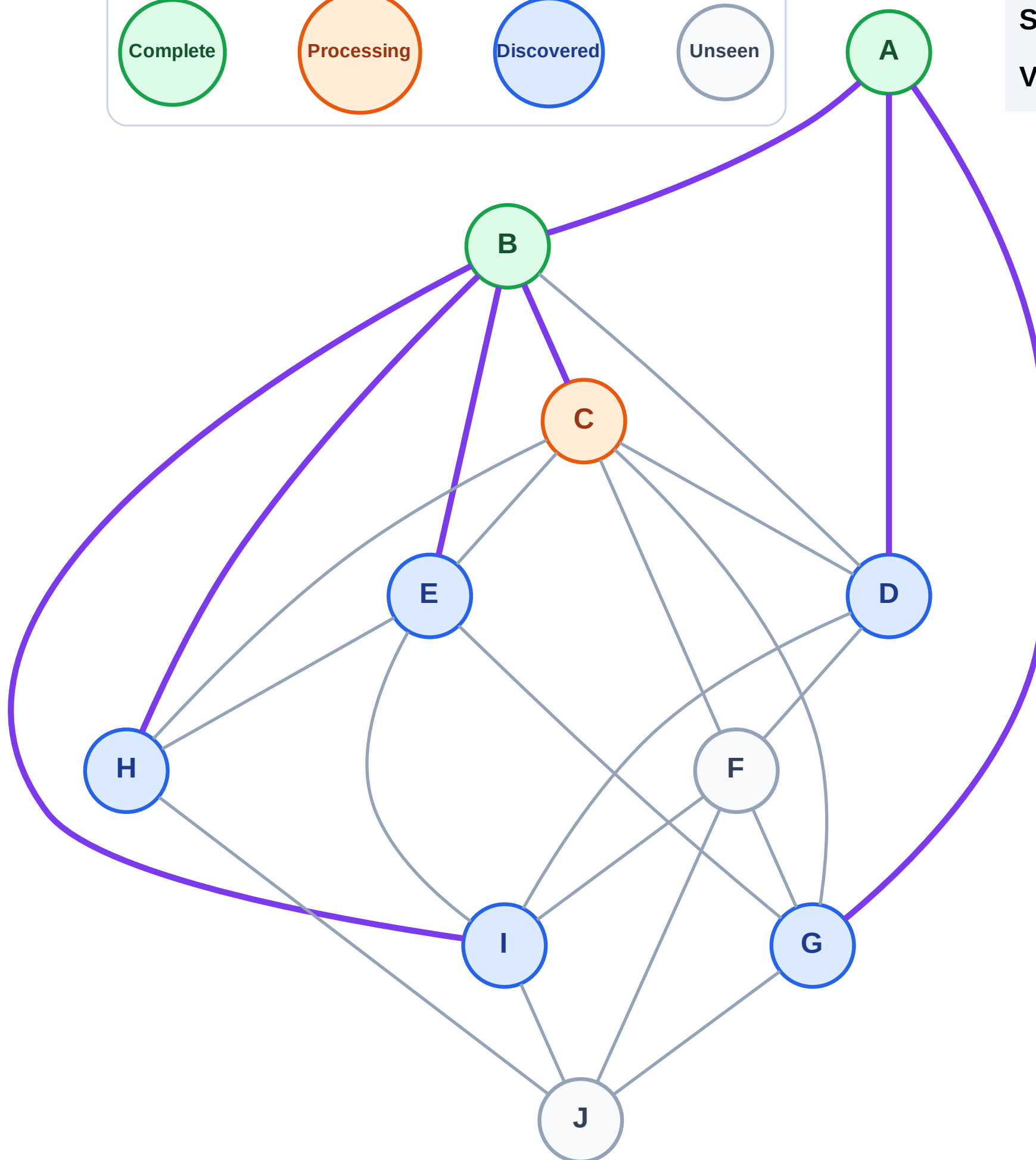
Step 6: Process node C

Legend



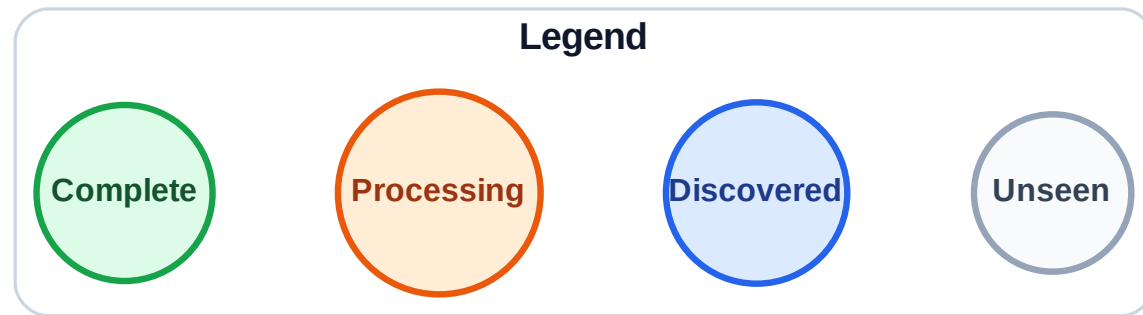
Stack (top at right): G | D | I | H | E

Visited: A, B



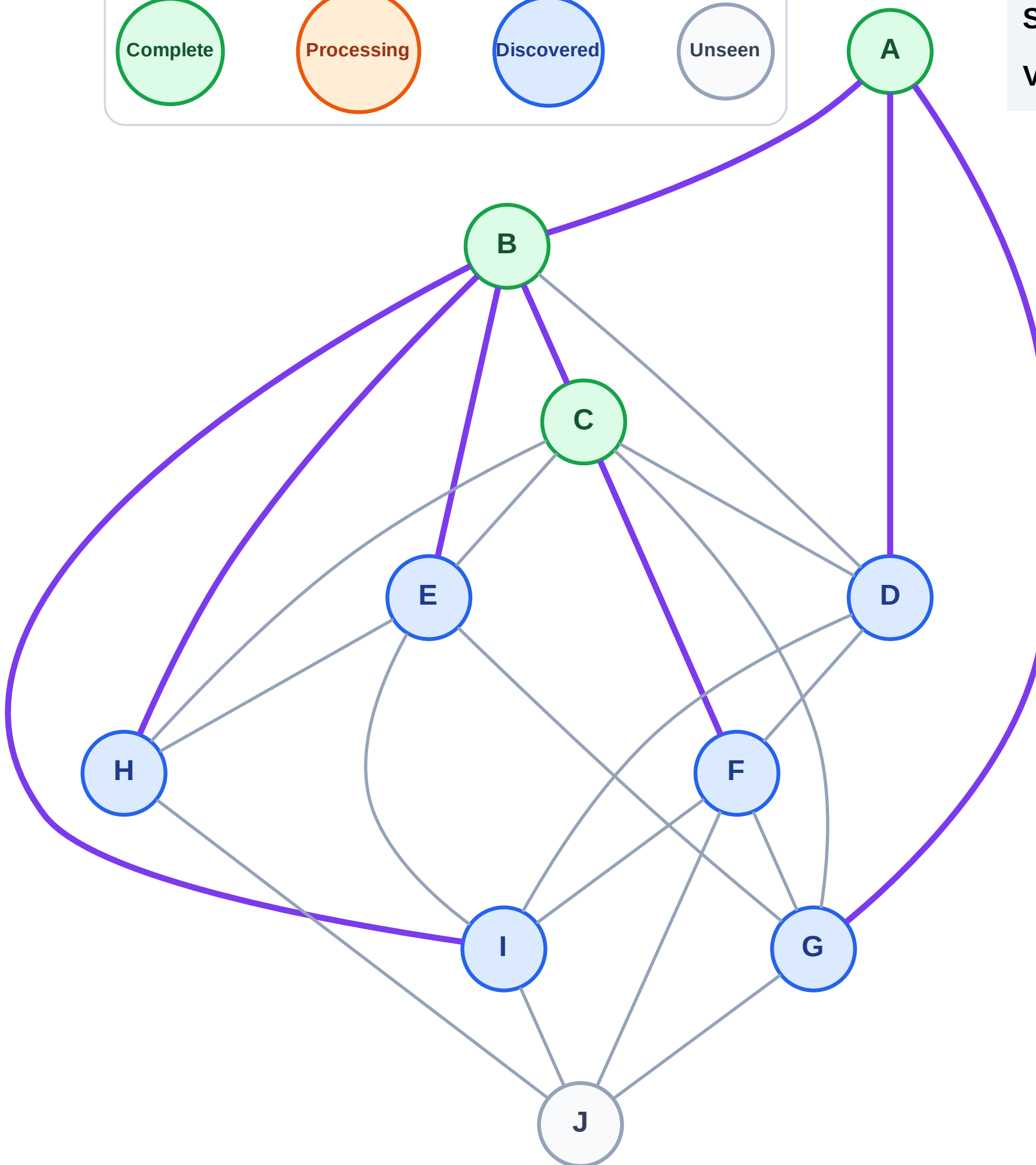
DFS Traversal

Step 7: Discover F from C



Stack (top at right): G | D | I | H | E | F

Visited: A, B, C



DFS Traversal

Step 8: Process node F

Legend

Complete

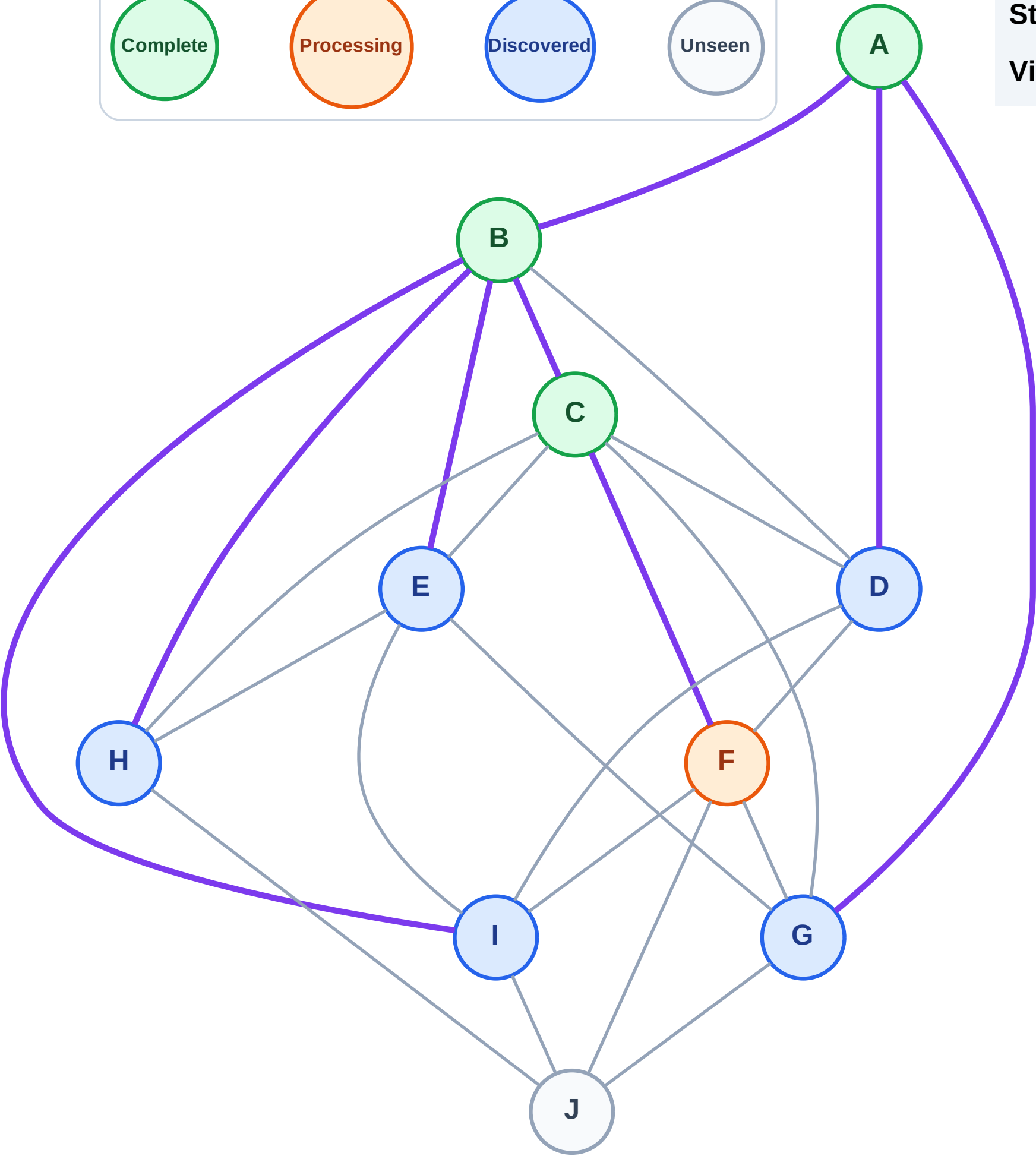
Processing

Discovered

Unseen

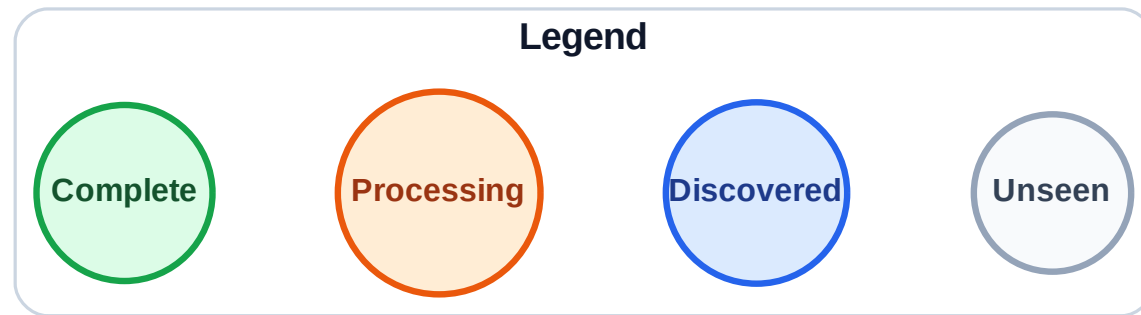
Stack (top at right): G | D | I | H | E

Visited: A, B, C



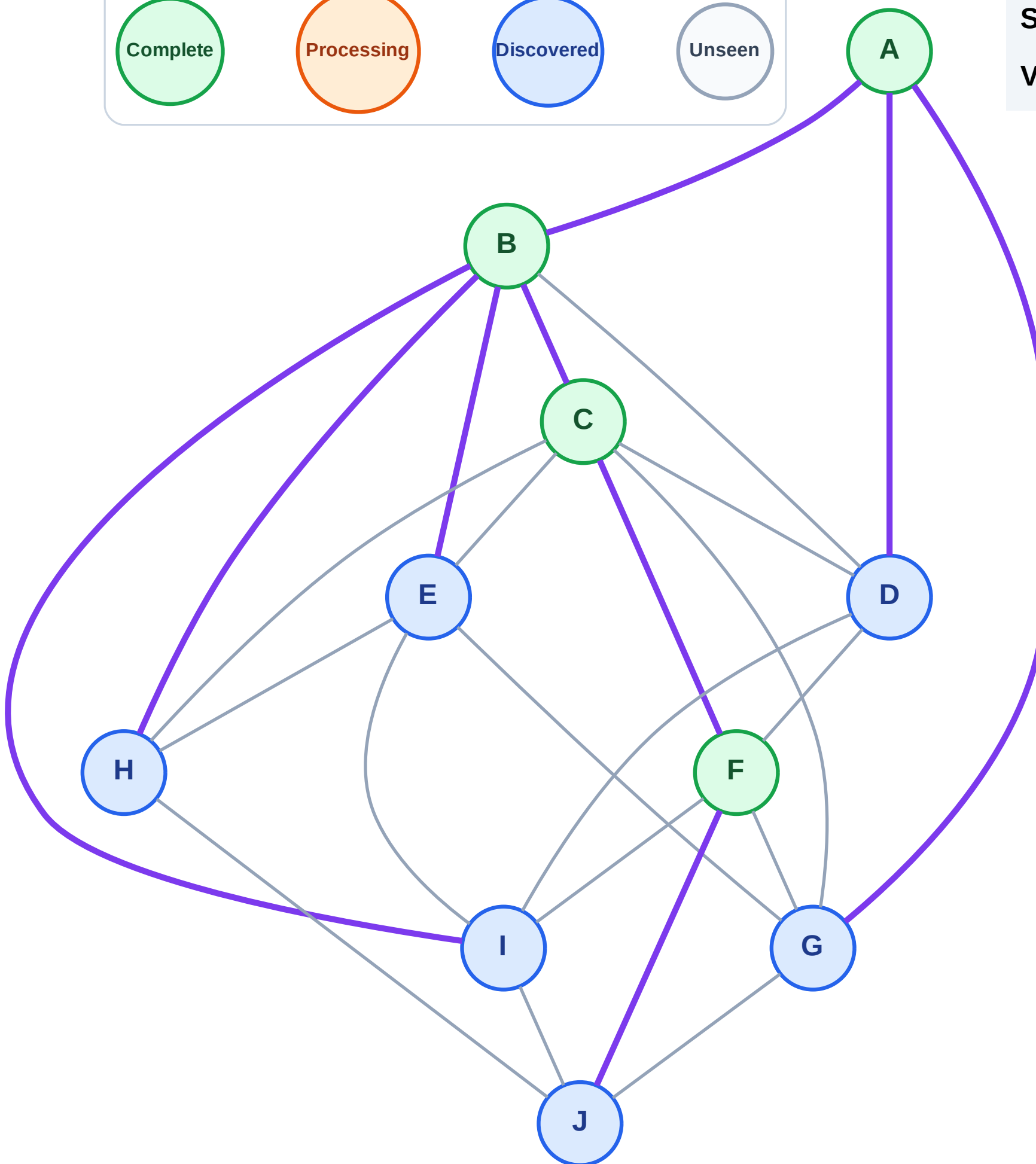
DFS Traversal

Step 9: Discover J from F



Stack (top at right): G | D | I | H | E | J

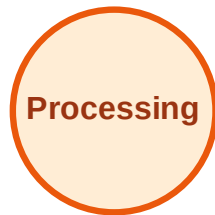
Visited: A, B, C, F



DFS Traversal

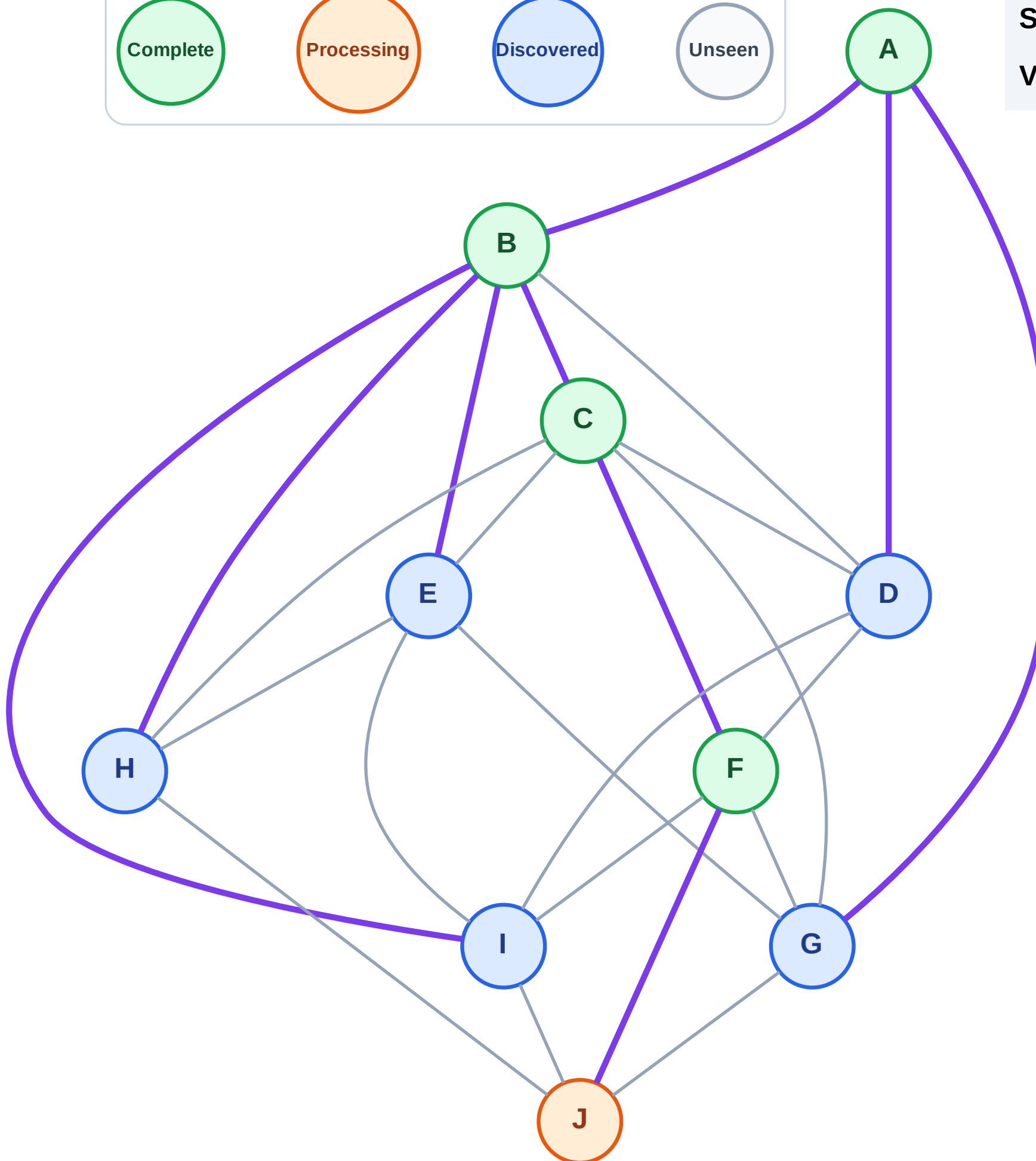
Step 10: Process node J

Legend



Stack (top at right): G | D | I | H | E

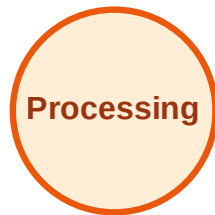
Visited: A, B, C, F



DFS Traversal

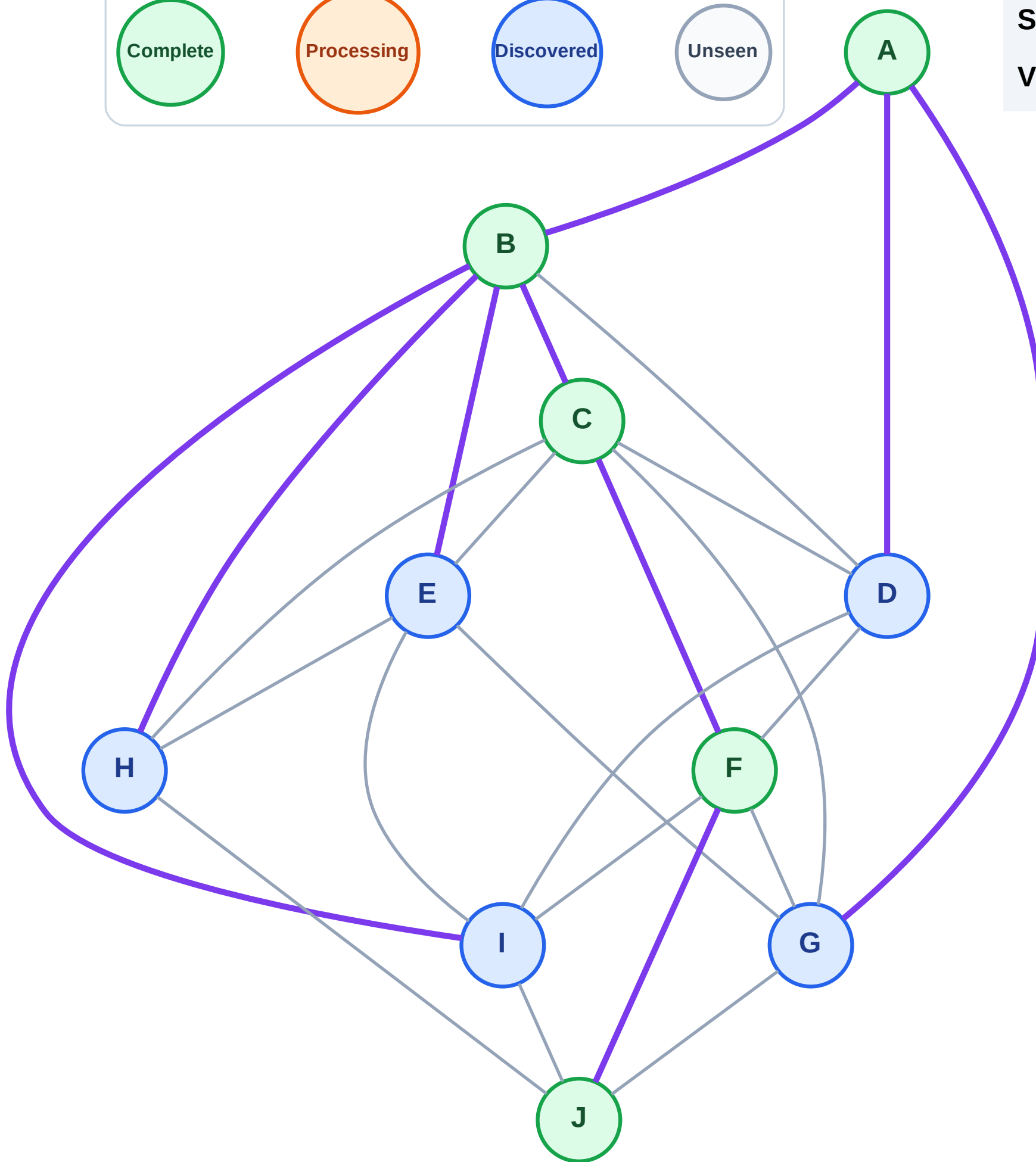
Step 11: No new neighbors from J

Legend



Stack (top at right): G | D | I | H | E

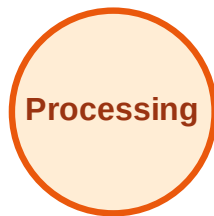
Visited: A, B, C, F, J



DFS Traversal

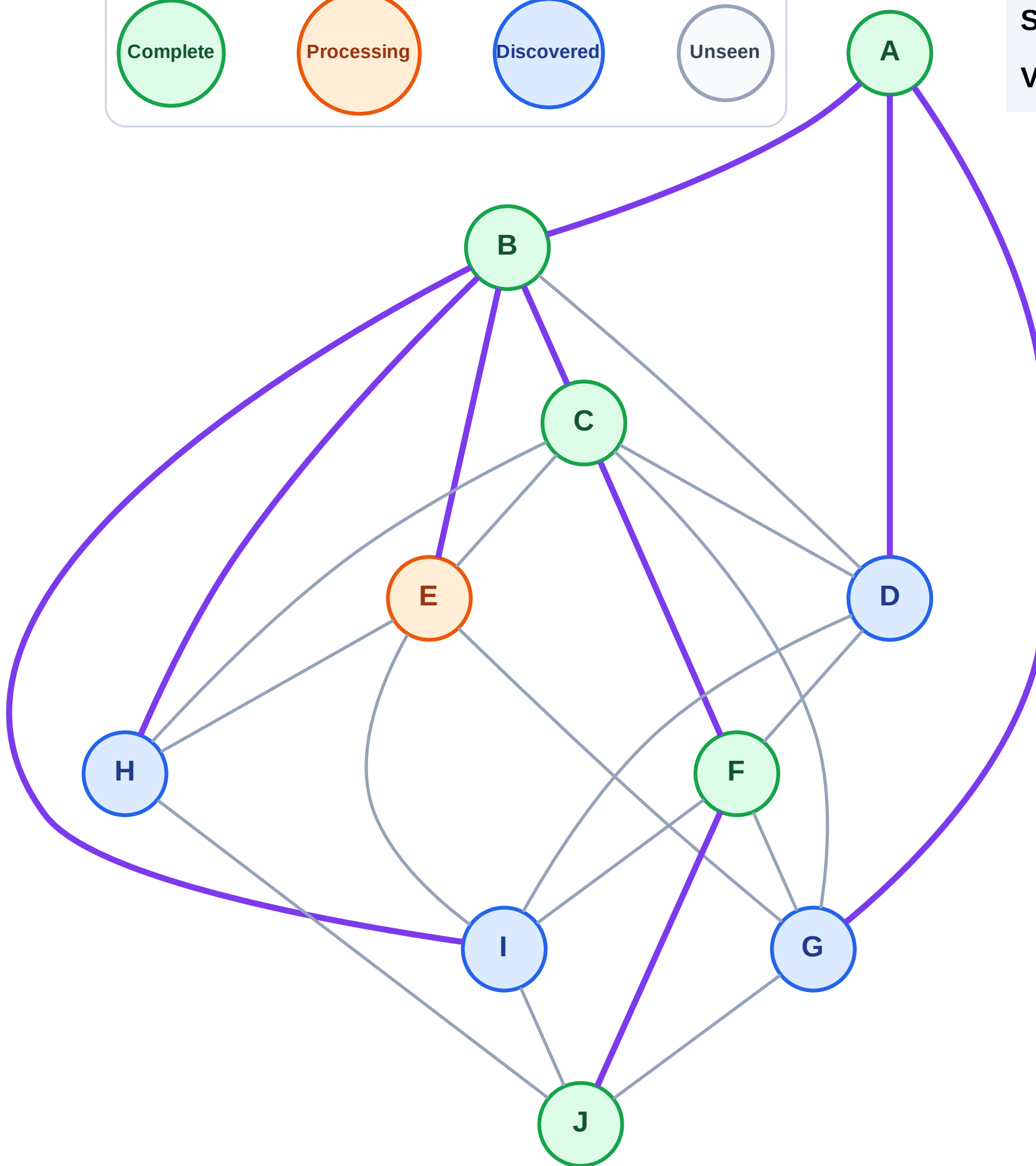
Step 12: Process node E

Legend



Stack (top at right): G | D | I | H

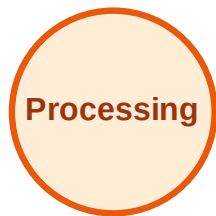
Visited: A, B, C, F, J



DFS Traversal

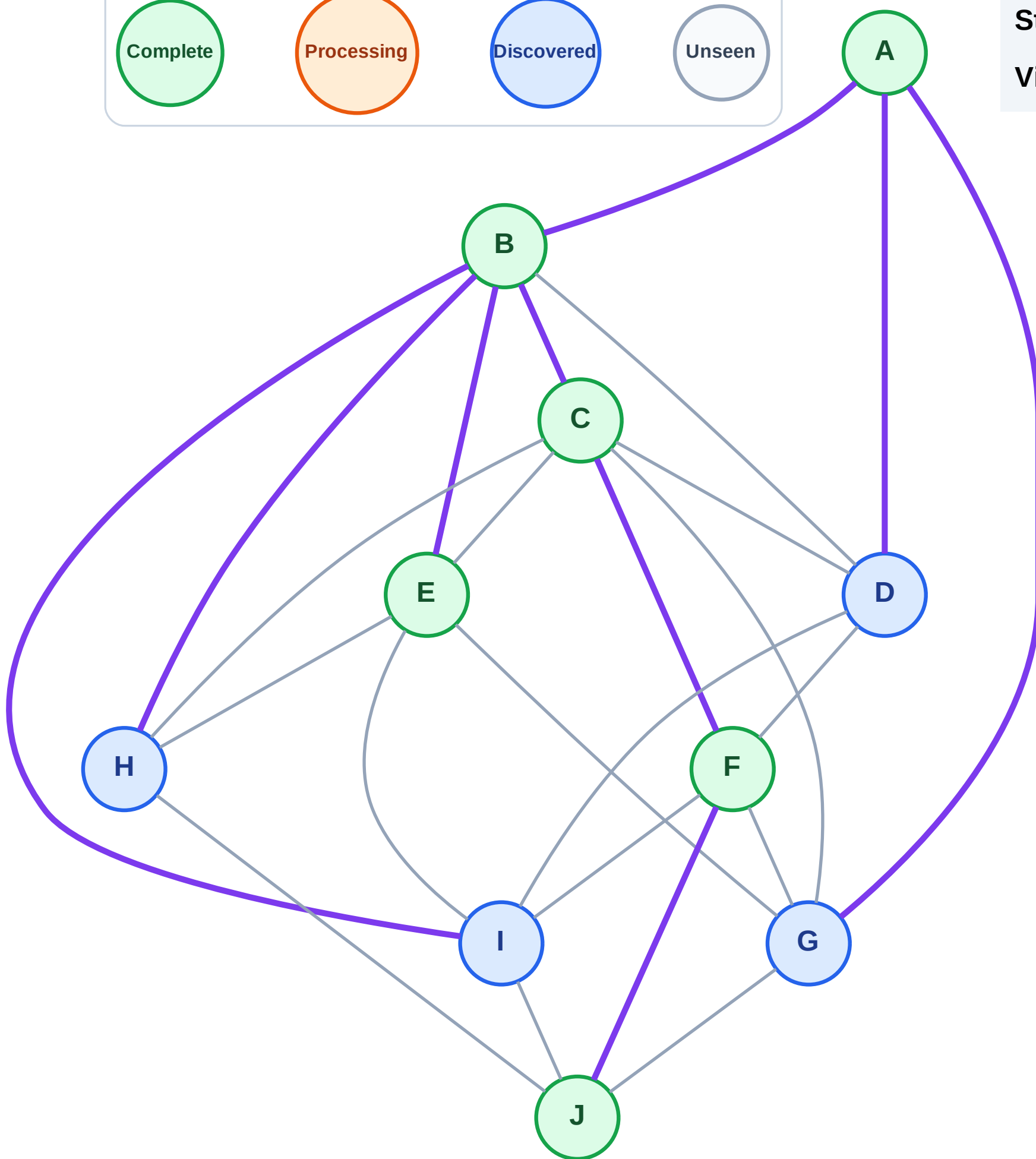
Step 13: No new neighbors from E

Legend



Stack (top at right): G | D | I | H

Visited: A, B, C, E, F, J



DFS Traversal

Step 14: Process node H

Legend

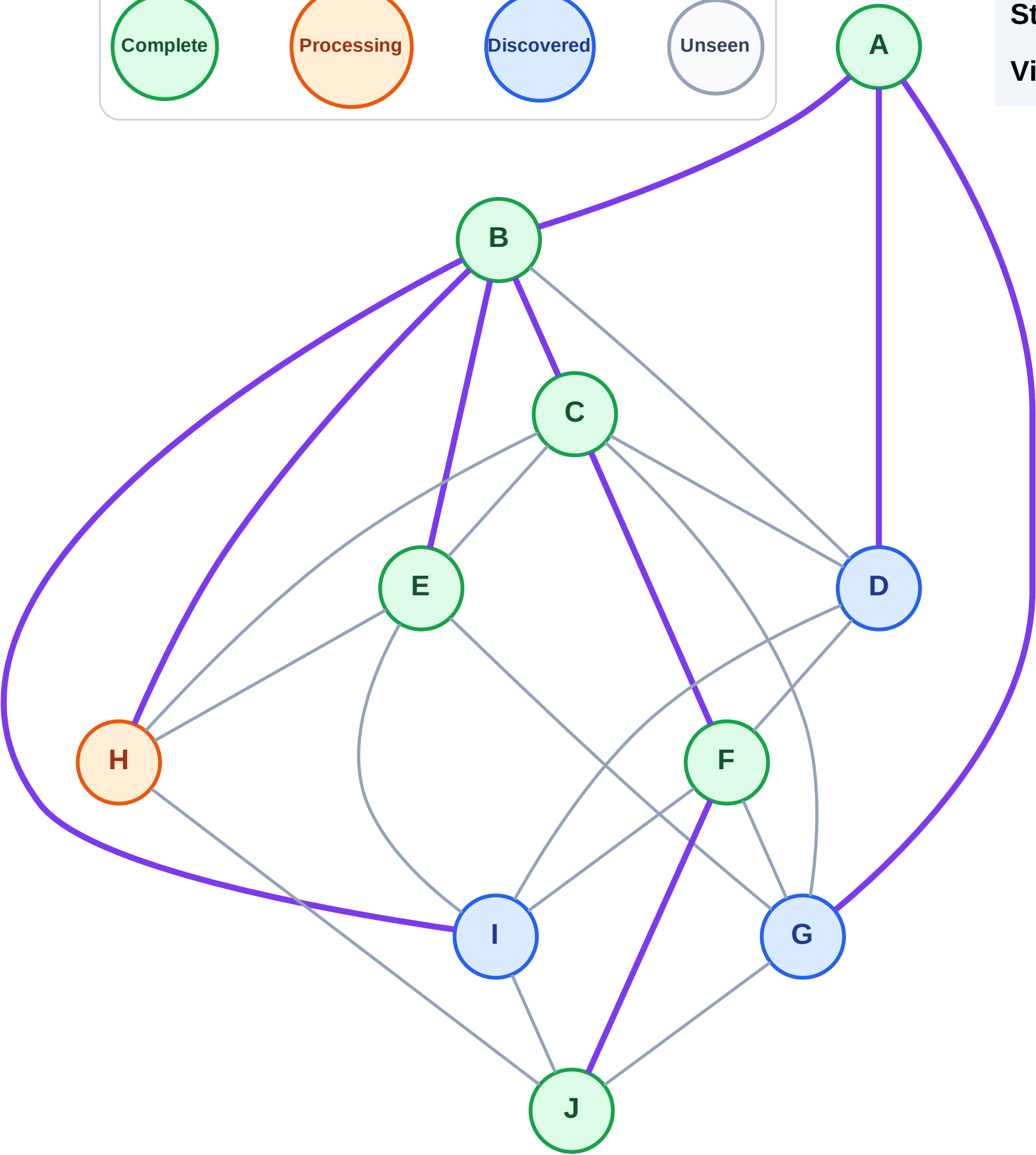
Complete

Processing

Discovered

Unseen

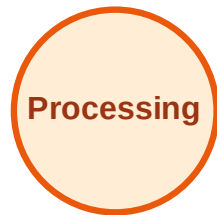
Stack (top at right): G | D | I
Visited: A, B, C, E, F, J



DFS Traversal

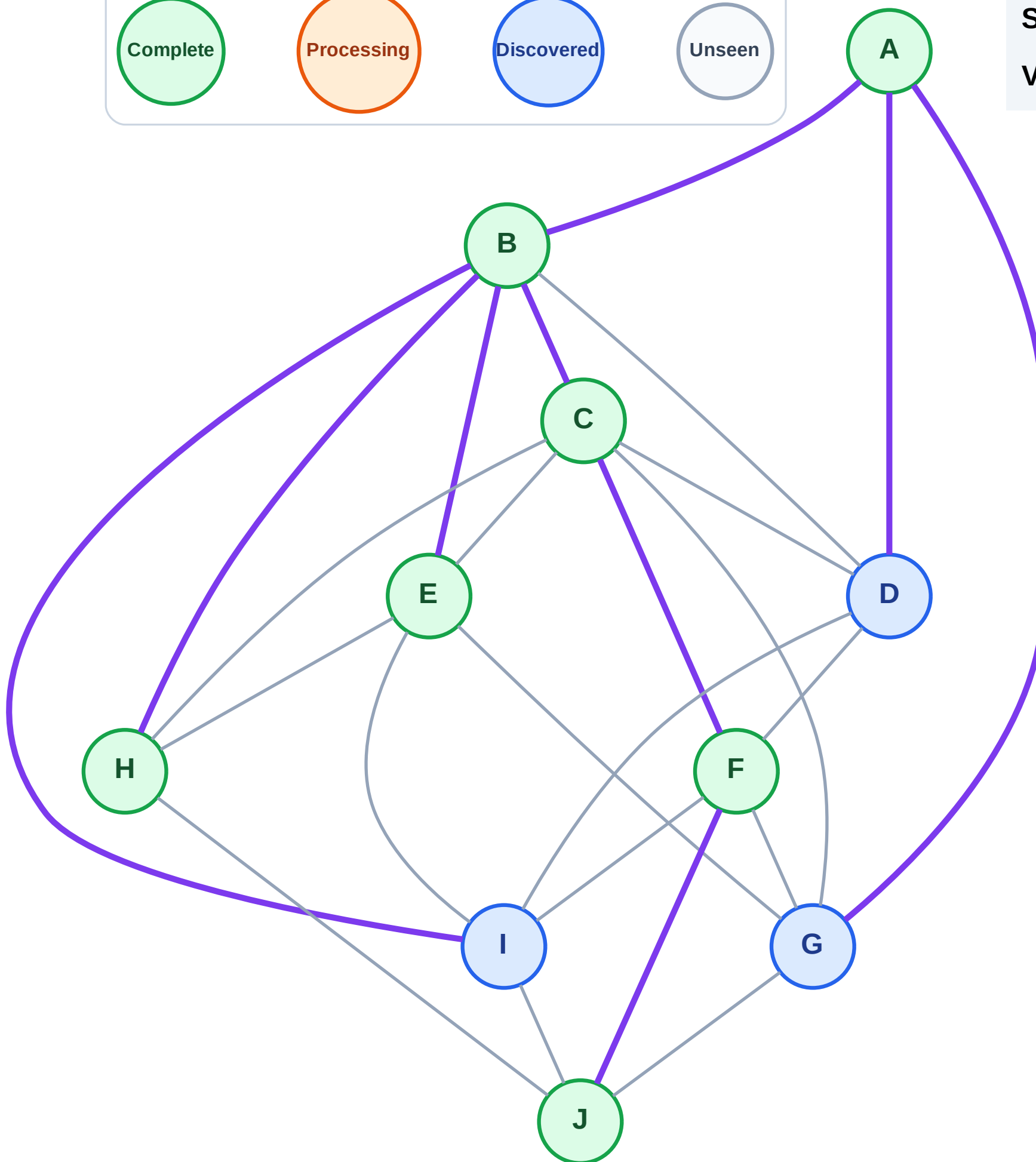
Step 15: No new neighbors from H

Legend



Stack (top at right): G | D | I

Visited: A, B, C, E, F, H, J



DFS Traversal

Step 16: Process node I

Legend

Complete

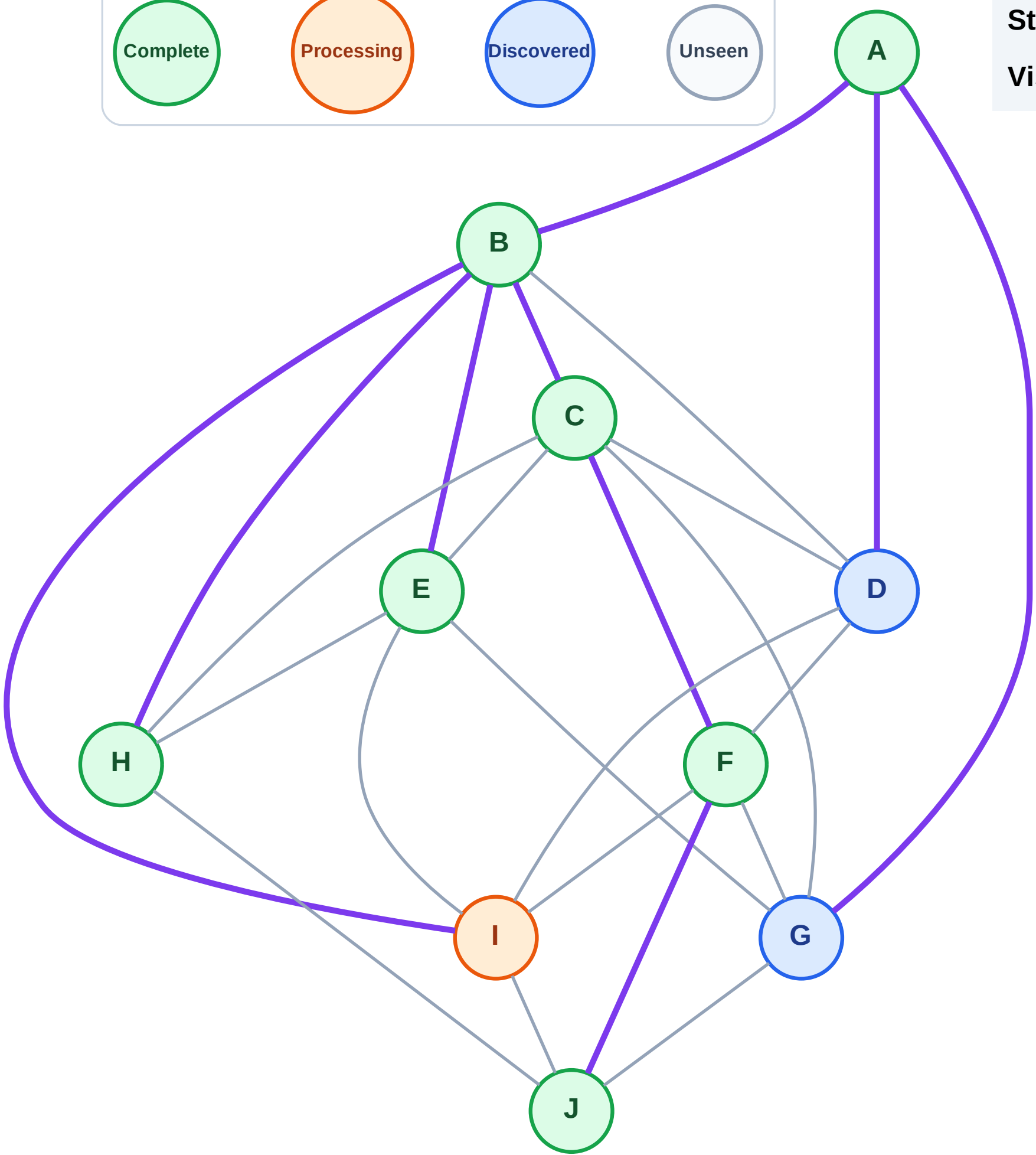
Processing

Discovered

Unseen

Stack (top at right): G | D

Visited: A, B, C, E, F, H, J



DFS Traversal

Step 17: No new neighbors from I

Legend

Complete

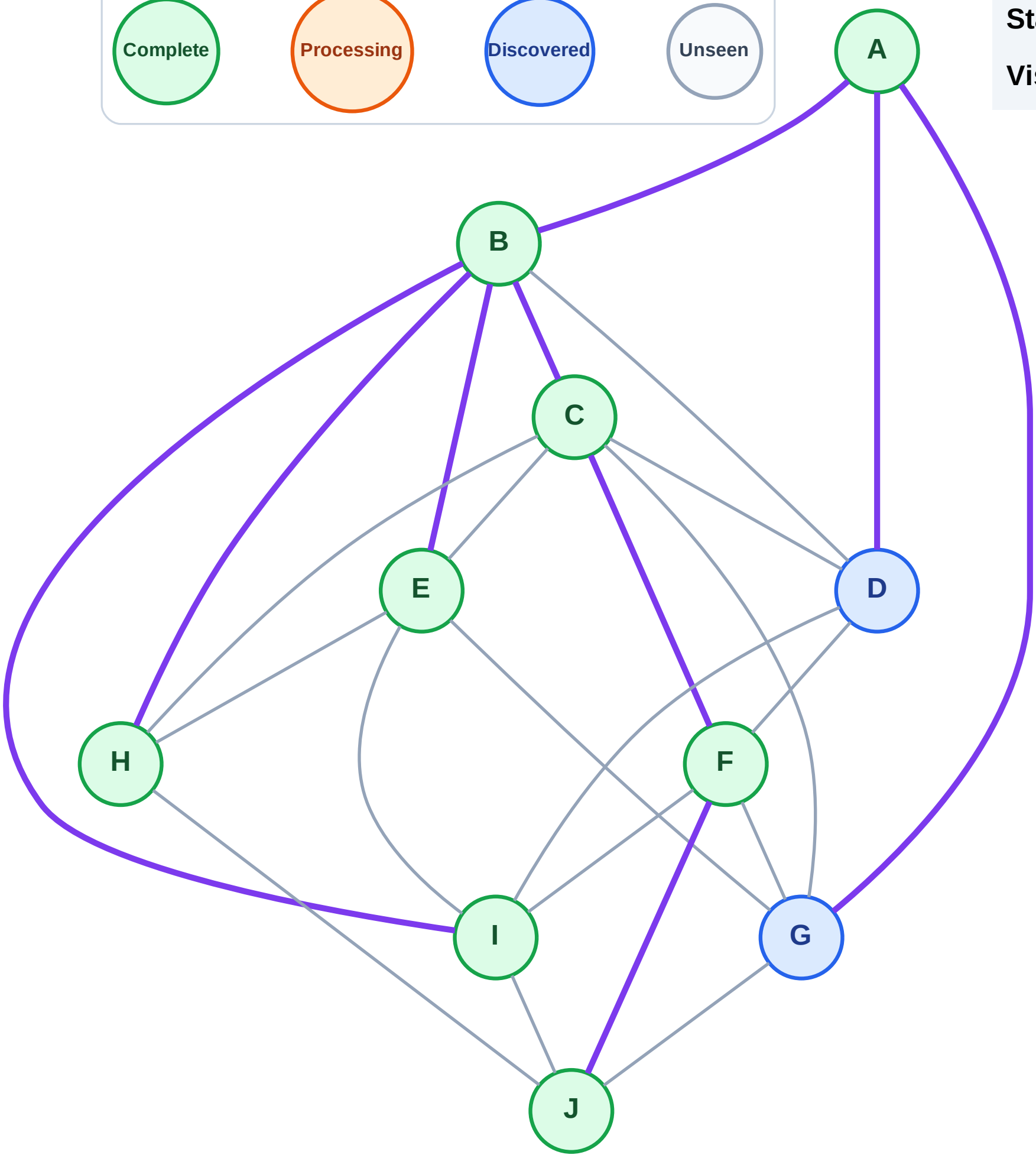
Processing

Discovered

Unseen

Stack (top at right): G | D

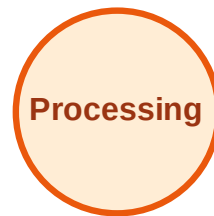
Visited: A, B, C, E, F, H, I, J



DFS Traversal

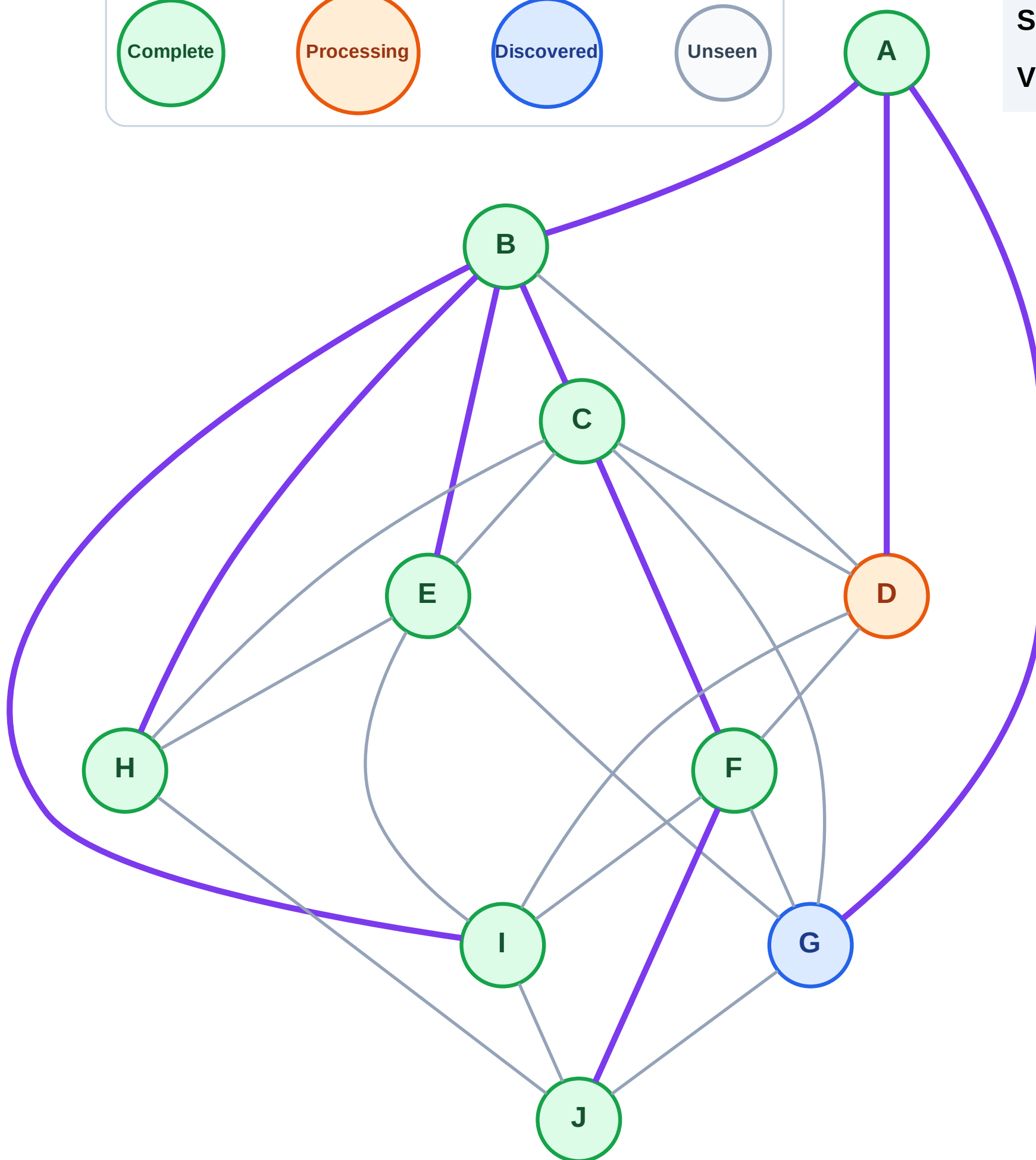
Step 18: Process node D

Legend



Stack (top at right): G

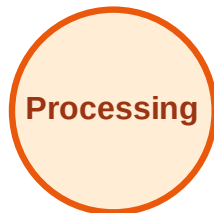
Visited: A, B, C, E, F, H, I, J



DFS Traversal

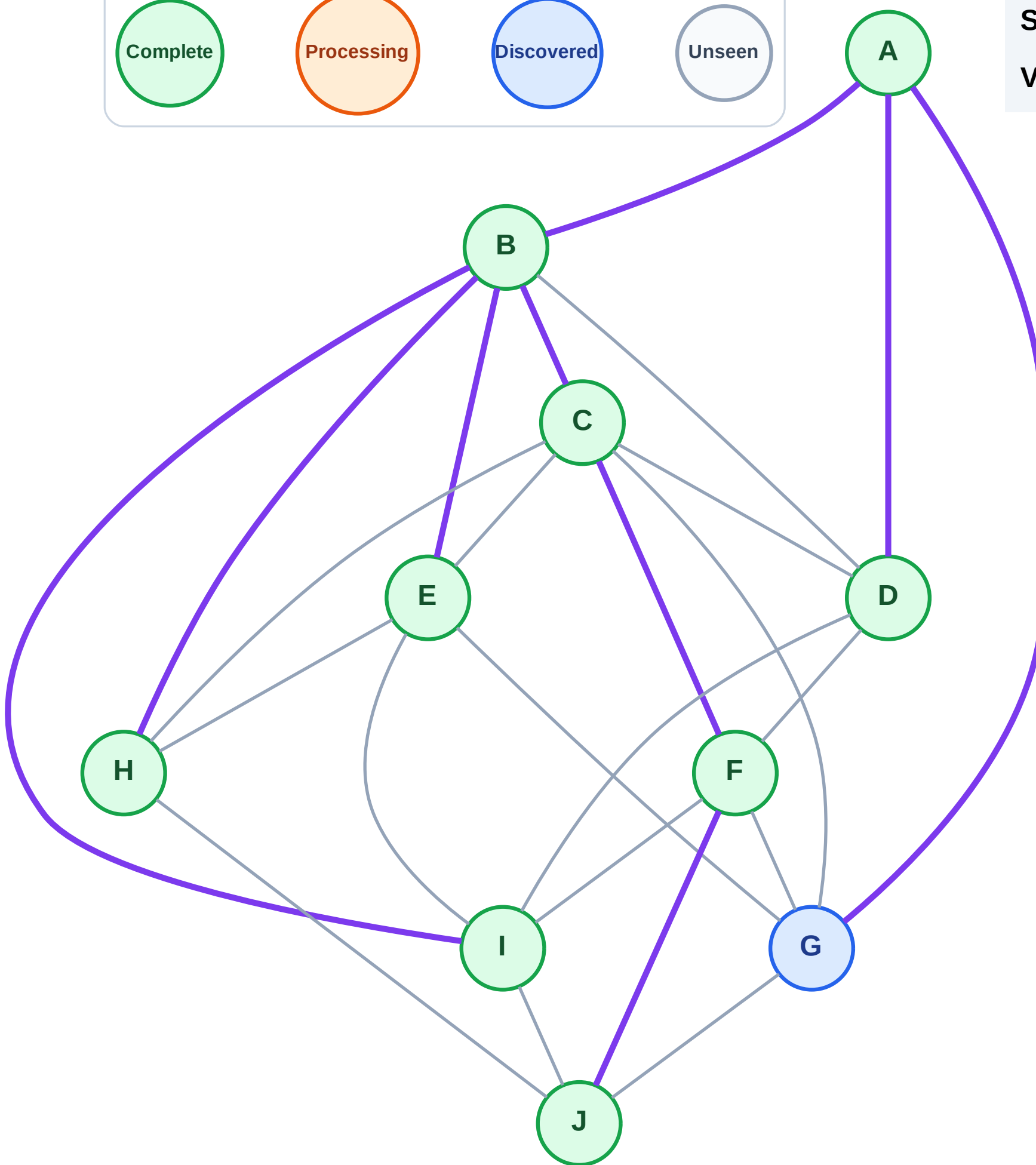
Step 19: No new neighbors from D

Legend



Stack (top at right): G

Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

Step 20: Process node G

Legend

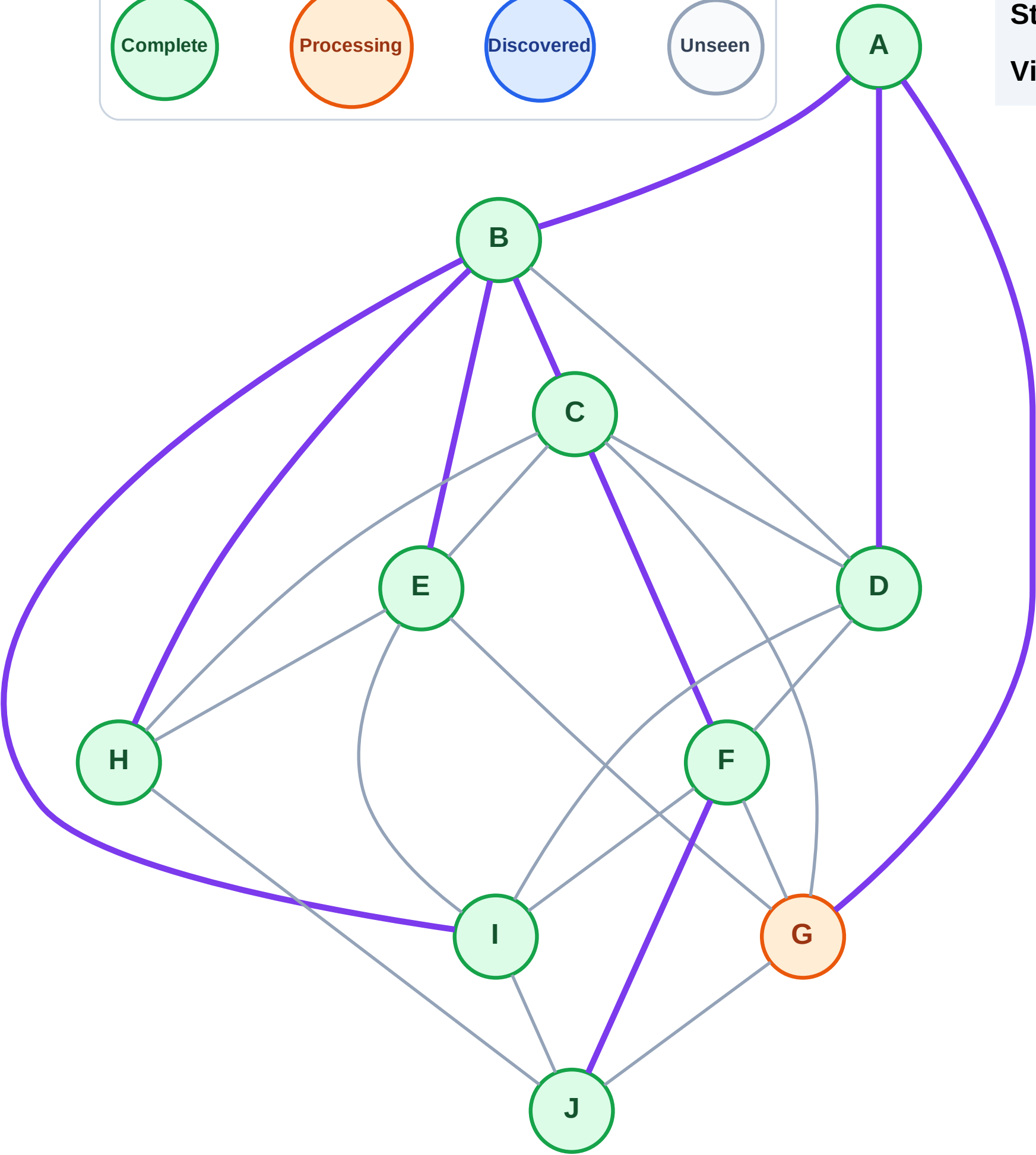
Complete

Processing

Discovered

Unseen

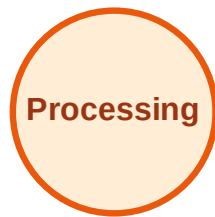
Stack (top at right): (empty)
Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

Step 21: No new neighbors from G

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

